Notes on Homological Algebra

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ABSTRACT: An edited transcription of the author's 34-page handwritten notes, following their route from additive and abelian categories through homology, chain homotopy, resolutions, derived functors, and adjunctions. Proof sketches and diagrams are reconstructed from the manuscript; substantive corrections and supplementary explanations are explicitly marked.

KEYWORDS: homological algebra, exact sequences, resolutions, derived functors

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About this edition

This edition follows the order and mathematical intentions of the 34-page handwritten manuscript. The manuscript is the primary source; the supplied MinerU transcription is used only as a recognition aid. Small handwritten diagrams have been redrawn as vector diagrams, with their objects, arrows and commutative relations checked against the surrounding argument. References to "manuscript page" always mean a page of the original PDF, not the present pagination.

The labels *Editorial correction*, *Editorial clarification*, and *Supplementary example* distinguish substantive repairs or additions from the author's exposition. An unfinished argument remains identified as such. The inserted textbook statements and exercises in the manuscript are attributed to Weibel [Wei94]; the book is also used to check conventions and supply limited context. The source-to-text ledger and a diagram audit accompany the editable project.

1 Additive and abelian categories

Source route. This section follows handwritten pages 1–4: first the additive structure on morphisms, then universal constructions, and finally the axioms needed for homology. A category $\mathcal{A}$ has objects, sets of morphisms $\mathrm{Hom}_{\mathcal{A}}(A, B)$, associative composition, and identity morphisms. We write $gf = g \circ f$, so the rightmost map acts first. A category is *preadditive*, or an *Ab-category*, when each morphism set is an abelian group and composition is additive in each variable. Thus, for compatible maps, $h(g' + g)f = hg'f + hgf$. The zero in each morphism group is called a zero morphism.

An *initial* object I has exactly one morphism $I \to X$ for every X; a *terminal* object T has exactly one morphism $X \to T$. A *zero object* is both, and is denoted by 0. The composite $X \to 0 \to Y$ is the zero morphism in a preadditive category. These objects, when they exist, are unique up to unique isomorphism; this sharpens the handwritten phrase "unique up to equivalence." In Set the empty set is initial and a singleton is terminal, so there is no zero object and this category is not additive.

A product of A, B is an object P with projections $p_1 : P \to A$, $p_2 : P \to B$ such that every pair $f : X \to A$, $g : X \to B$ factors uniquely as $p_1\langle f, g\rangle = f$, $p_2\langle f, g\rangle = g$. A coproduct reverses these arrows: it has injections $i_1 : A \to Q$, $i_2 : B \to Q$, and every pair $f : A \to X$, $g : B \to X$ determines a unique $[f, g] : Q \to X$. Figure 1 redraws the two small universal diagrams in the manuscript. Cartesian product and disjoint union give the corresponding constructions in sets; these examples are pictures of the universal properties, rather than definitions valid in every category.

Finite biproducts. An additive category is a preadditive category with a zero object and all binary products, hence all finite products. The manuscript proves that these

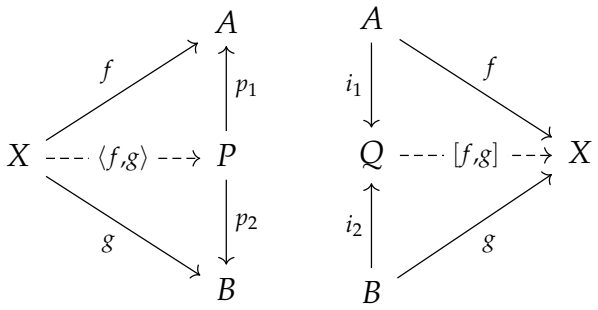


Figure 1. Product and coproduct factorization. Every triangular path agrees with its direct arrow, and the dashed mediating morphism is unique.

products are also coproducts. Indeed, the product property constructs $i_1 : A \to P$, $i_2 : B \to P$ from $(\mathrm{id}_A, 0)$ and $(0, \mathrm{id}_B)$. Hence $p_j i_i$ is the identity for $i = j$ and zero otherwise. Put $e = i_1 p_1 + i_2 p_2$. Since $p_1 e = p_1$ and $p_2 e = p_2$, uniqueness for the product gives

$$p_j i_i = \delta_{ij}\,\mathrm{id}, \qquad i_1 p_1 + i_2 p_2 = \mathrm{id}_P\,. \tag{1.1}$$

For maps $f : A \to X$, $g : B \to X$, the morphism $f p_1 + g p_2$ has the required restrictions to A and B. If h has the same restrictions, then $h = h(i_1 p_1 + i_2 p_2) = f p_1 + g p_2$, which proves uniqueness. The reversed argument starts from a coproduct. We therefore write $P = A \oplus B$ and call it a *biproduct*. The notation $\bigoplus_{\alpha \in I} A_\alpha$ for an arbitrary coproduct does not assert that it is also a product: the proof uses a finite sum of morphisms, whereas an abelian group carries no automatic infinite-summation operation. For example, the direct sum of infinitely many copies of $\mathbb{Z}$ consists of finitely supported sequences and differs from their direct product.

Modules and complexes. The category Mod-R of unital right modules over a unital ring R is additive. The same holds for its category of chain complexes. Here a chain complex C consists of modules C_n and differentials $d_n : C_n \to C_{n-1}$, indexed by $n \in \mathbb{Z}$, with $d_{n-1} d_n = 0$. A chain map $f : C \to D$ has components $f_n : C_n \to D_n$ satisfying $d_D f = f d_C$. Set $Z_n(C) = \ker d_n$, $B_n(C) = \operatorname{im} d_{n+1}$, and $H_n(C) = Z_n(C)/B_n(C)$. The first two are the cycles and boundaries; homology identifies cycles differing by a boundary. Products and coproducts of module complexes are constructed degreewise, with differentials $\prod_\alpha d_\alpha$ and $\bigoplus_\alpha d_\alpha$; the zero complex is the zero object.

Exercise retained: Weibel 1.2.1. Show that the canonical maps identify $\bigoplus_\alpha H_n(C_\alpha)$ with $H_n(\bigoplus_\alpha C_\alpha)$, and $\prod_\alpha H_n(C_\alpha)$ with $H_n(\prod_\alpha C_\alpha)$, for module complexes. *Editorial bridge.* Cycles are computed componentwise. Boundaries are also componentwise: finitely supported boundaries admit finitely supported lifts, and a family of boundaries admits a family of lifts using the usual axiom of choice. This proves the assertions by passing to quotients. The product assertion is being made for modules; it is not an

assertion that arbitrary products are exact in every abelian category.

1.1 Kernels, cokernels, and cancellation

For $f : B \to C$ in an additive category, a kernel is a map $i : A \to B$ with $fi = 0$ such that every $g : A' \to B$ satisfying $fg = 0$ factors uniquely through i. A cokernel is a map $e : C \to D$ with $ef = 0$ such that every $g : C \to D'$ satisfying $gf = 0$ factors uniquely through e. The relevant arrows, including their domains, are shown in Figure 2. The symbols $\ker f$ and $\operatorname{coker} f$ sometimes denote the objects and sometimes their structural morphisms; we specify the map when the distinction matters.

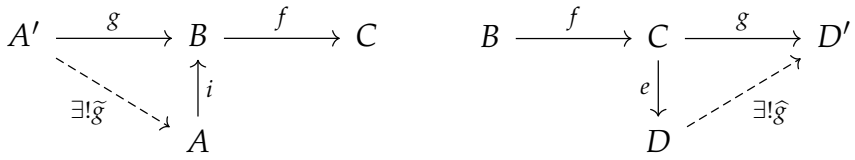


Figure 2. The universal factorizations defining a kernel and a cokernel. The left diagram assumes $fg = 0$, and the right diagram assumes $gf = 0$.

Monomorphisms and epimorphisms. A morphism $i : A \to B$ is *monic* when $ig_1 = ig_2$ implies $g_1 = g_2$ for every pair $g_1, g_2 : X \to A$. Dually, $e : C \to D$ is *epic* when $g_1 e = g_2 e$ implies $g_1 = g_2$ for every pair $g_1, g_2 : D \to X$. In an additive category subtraction gives the equivalent tests $ig = 0 \Rightarrow g = 0$ and $ge = 0 \Rightarrow g = 0$. Every kernel is monic: if $ig_1 = ig_2 = h$, then $fh = 0$, and g_1, g_2 are two factorizations of the same h through its kernel, so uniqueness gives equality. Reversing this argument proves that every cokernel is epic.

An epimorphism need not be surjective on underlying sets. The manuscript uses the inclusion $\mathbb{Z} \hookrightarrow \mathbb{Q}$ in the category of unital rings. If unital ring maps $g, h : \mathbb{Q} \to S$ agree on $\mathbb{Z}$, then, for $m \neq 0$, both send m^{-1} to the inverse of the common image of m. Hence $g(n/m) = h(n/m)$, proving right cancellation. The inclusion is therefore epic, although it is not surjective as a function. The category of unital rings is not additive: the pointwise sum of two ring homomorphisms need not preserve multiplication or the identity. In fact, $(f+g)(ab) = f(a)f(b) + g(a)g(b)$, whereas $(f+g)(a)(f+g)(b)$ also contains the cross terms $f(a)g(b) + g(a)f(b)$. This explains why module intuition cannot simply be transferred to rings; nonadditivity alone, however, is not a general criterion for deciding whether epimorphisms are surjective.

1.2 Image, coimage, and the abelian condition

A *preabelian* category is an additive category in which every morphism has a kernel and a cokernel. For $f : A \to B$ define its image as the kernel of its cokernel, and its coimage

as the cokernel of its kernel. Thus $i : \operatorname{im} f \to B$ is a subobject, whereas $e : A \to \operatorname{coim} f$ is a quotient map. In particular, the coimage is not generally a subobject of A; the subset-like notation in the handwritten line is replaced here by its correct quotient interpretation.

The canonical comparison in Figure 3 is constructed in two stages, exactly as in the manuscript. Writing $q = \operatorname{coker} f$, the identity $qf = 0$ gives a unique $f_1 : A \to \operatorname{im} f$ with $if_1 = f$. If $k : \ker f \to A$ is the kernel map, then $if_1k = fk = 0$; since i is monic, $f_1k = 0$. The cokernel property of e gives a unique $\bar{f} : \operatorname{coim} f \to \operatorname{im} f$ satisfying $\bar{f}e = f_1$. Hence $f = i\bar{f}e$. Uniqueness uses cancellation by i and e.

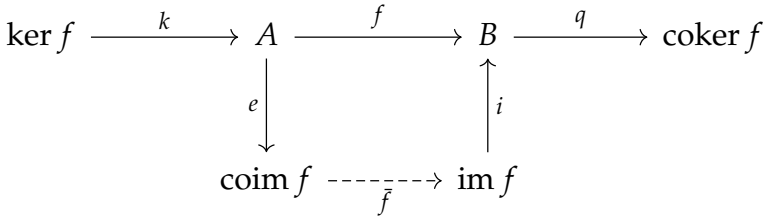


Figure 3. Canonical factorization in a preabelian category. The dashed map exists uniquely, but is not automatically an isomorphism without the abelian hypothesis.

The abelian axioms. An abelian category is a preabelian category in which every monomorphism is a kernel of its cokernel, and every epimorphism is a cokernel of its kernel. These assertions mean the corresponding universal properties, up to their canonical isomorphisms, rather than literal equality between differently presented objects. Equivalently, the canonical comparison $\bar{f}$ is an isomorphism for every f. This is the setting in which images, quotients, and exactness have the familiar module behavior. A composable pair $A \xrightarrow{u} B \xrightarrow{v} C$ is exact at B when $\operatorname{im} u = \ker v$ as subobjects of B.

2 The Snake lemma and long exact homology sequences

The goal of handwritten pages 5–9 is to explain the new map produced by a short exact sequence of complexes $0 \to A \xrightarrow{i} B \xrightarrow{\pi} C \to 0$. Exactness here is degreewise. The induced homology maps alone do not form a short exact sequence; a connecting map $\partial_n : H_n(C) \to H_{n-1}(A)$ supplies the missing information and yields a long exact sequence. We first reconstruct the Snake lemma that creates this map. The explicit element calculations below are proofs for modules or abelian groups. The categorical Snake lemma holds in every abelian category; using element notation in that generality requires the usual categorical diagram-chase justification, not an assumption that arbitrary objects literally have elements.

Consider Figure 4, whose rows are exact. No injectivity of α or surjectivity of p is initially assumed. The vertical identities are $g\alpha = if$ and $pg = h\pi$.

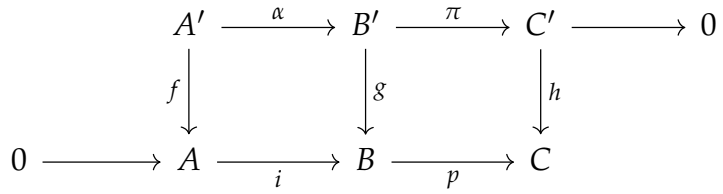


Figure 4. The Snake lemma input. Both rows are exact, and the two squares express $g\alpha = if$ and $pg = h\pi$.

Induced maps. The kernel maps are $\widetilde{\alpha} = \alpha|_{\ker f}$ and $\widetilde{\pi} = \pi|_{\ker g}$: for example, $g\alpha(x) = if(x) = 0$ when $x \in \ker f$. The cokernel maps are $\widetilde{i}([a]_f) = [i(a)]_g$ and $\widetilde{p}([b]_g) = [p(b)]_h$. Changing a by $f(a')$ changes $i(a)$ by $g\alpha(a')$, so the first quotient map is well-defined; the second follows from $pg = h\pi$. Here $[a]_f$ means the class of a in $\operatorname{coker} f = A/\operatorname{im} f$. The Snake lemma supplies the middle map in

$$\ker f \xrightarrow{\widetilde{\alpha}} \ker g \xrightarrow{\widetilde{\pi}} \ker h \xrightarrow{\partial} \operatorname{coker} f \xrightarrow{\widetilde{i}} \operatorname{coker} g \xrightarrow{\widetilde{p}} \operatorname{coker} h. \tag{2.1}$$

Constructing the connecting map. Take $c' \in \ker h$. Surjectivity of π permits a lift $b' \in B'$ with $\pi(b') = c'$. Then $pg(b') = h(c') = 0$, so exactness of the bottom row gives a unique $a \in A$ with $i(a) = g(b')$. Define $\partial(c') = [a]_f$. This is the precise meaning of the handwritten mnemonic $i^{-1}g\pi^{-1}$: π^{-1} means a choice of lift, and i^{-1} is used only on $\operatorname{im} i$.

If a second lift differs by $\alpha(a')$, its resulting a differs by $f(a')$, because $g\alpha = if$ and i is injective. Its class in the cokernel is unchanged. Additivity follows by adding lifts, so ∂ is a homomorphism. The quotient by $\operatorname{im} f$ removes exactly the ambiguity in the lift.

Checking all four interior terms. At $\ker g$, the composite $\widetilde{\pi}\widetilde{\alpha}$ is zero. Conversely, if $b' \in \ker g$ and $\pi(b') = 0$, write $b' = \alpha(a')$. Then $if(a') = g(b') = 0$, and injectivity of i gives $a' \in \ker f$.

At $\ker h$, a lift already lying in $\ker g$ gives $a = 0$, so $\partial\widetilde{\pi} = 0$. Conversely, if $\partial(c') = 0$, a chosen lift b' satisfies $g(b') = i(a)$ with $a = f(a')$. Correct the lift to $\widehat{b}' = b' - \alpha(a')$. Now $g(\widehat{b}') = 0$ while $\pi(\widehat{b}') = c'$. This correction is the essential bridge from a zero cokernel class to an actual kernel element.

At $\operatorname{coker} f$, a class in the image of ∂ satisfies $i(a) = g(b')$, so its image in $\operatorname{coker} g$ vanishes. Conversely, if $[i(a)]_g = 0$, choose b' with $i(a) = g(b')$. Then $c' = \pi(b')$ lies in $\ker h$ because $h(c') = pg(b') = pi(a) = 0$, and $\partial(c') = [a]_f$.

At $\operatorname{coker} g$, the composite $\widetilde{p}\widetilde{i}$ is zero. If $\widetilde{p}([b]_g) = 0$, there is $c' \in C'$ with $p(b) = h(c')$. Lift c' to $b' \in B'$. Then $p(b - g(b')) = 0$, so $b - g(b') = i(a)$ for some a. Thus $[b]_g = [i(a)]_g$, proving the remaining equality of image and kernel.

Endpoint enhancements. If α is monic, its restriction $\widetilde{\alpha}$ is monic. If p is epic, its induced quotient map $\widetilde{p}$ is epic. These are additional hypotheses for adding zeros to the ends of (2.1); they are not part of the basic six-term statement. The construction is natural under a morphism of such diagrams: applying the morphism to a chosen lift gives a lift in the new diagram, and both routes give the same cokernel class.

2.1 Applying the Snake lemma to homology

For a complex C, the differential descends to $\bar{d}_n : C_n/B_n(C) \to Z_{n-1}(C)$, $[c] \mapsto d_n c$. It is well-defined because $d^2 = 0$. Its kernel is $H_n(C)$ and its cokernel is $H_{n-1}(C)$. Consequently,

$$0 \longrightarrow H_n(C) \longrightarrow C_n/B_n(C) \xrightarrow{\bar{d}_n} Z_{n-1}(C) \longrightarrow H_{n-1}(C) \longrightarrow 0 \tag{2.2}$$

is exact.

Splicing. The elementary mechanism behind this four-term sequence is worth retaining. For any map $f : X \to Y$ in an abelian category, the two short exact sequences through $\operatorname{im} f$ splice to $0 \to \ker f \to X \xrightarrow{f} Y \to \operatorname{coker} f \to 0$. More generally, from $0 \to K \to X \xrightarrow{\rho} M \to 0$ and $0 \to M \xrightarrow{j} Y \to Q \to 0$, one obtains $0 \to K \to X \xrightarrow{j\rho} Y \to Q \to 0$. Injectivity of j gives $\ker(j\rho) = \ker \rho$, and surjectivity of ρ gives $\operatorname{im}(j\rho) = \operatorname{im} j$.

For the short exact sequence $0 \to A \xrightarrow{i} B \xrightarrow{\pi} C \to 0$, use Figure 5. Its top row is right exact and its bottom row is left exact. The two crossed-out endpoint zeros in the handwritten diagram are intentionally omitted here.

$$\begin{array}{ccccccc} & & A_n/B_n(A) & \xrightarrow{\bar{i}} & B_n/B_n(B) & \xrightarrow{\bar{\pi}} & C_n/B_n(C) \longrightarrow 0 \\ & & \downarrow{\scriptstyle \bar{d}_n^A} & & \downarrow{\scriptstyle \bar{d}_n^B} & & \downarrow{\scriptstyle \bar{d}_n^C} \\ 0 & \longrightarrow & Z_{n-1}(A) & \xrightarrow[i_Z]{} & Z_{n-1}(B) & \xrightarrow[\pi_Z]{} & Z_{n-1}(C) \end{array}$$

Figure 5. The quotient-and-cycle construction for the homology Snake lemma. No left endpoint zero is asserted for the top row, nor a right endpoint zero for the bottom row.

Editorial reconstruction of row exactness. The quotient map $\bar{\pi}$ is surjective because π_n is. If $\pi_n(b)$ is a boundary, choose c_{n+1} with $d_C c_{n+1} = \pi_n(b)$, and lift it to b_{n+1}. The element $b - d_B b_{n+1}$ lies in $\ker \pi_n = \operatorname{im} i_n$, proving exactness in the middle of the top row. For the bottom row, i_Z is injective. If a cycle b satisfies $\pi(b) = 0$, write $b = i(a)$;

then $i(da) = db = 0$, so a is a cycle. These checks supply the hypotheses needed for the Snake lemma, rather than merely assuming that quotient and cycle operations preserve short exactness.

By (2.2), the kernels and cokernels of the vertical maps are the corresponding homology groups. The Snake lemma therefore yields the successive pieces of

$$\cdots \longrightarrow H_n(A) \xrightarrow{i_*} H_n(B) \xrightarrow{\pi_*} H_n(C) \xrightarrow{\partial_n} H_{n-1}(A) \longrightarrow \cdots . \tag{2.3}$$

Explicitly, lift a cycle $c \in C_n$ to $b \in B_n$, write $d_B b = i(a)$, and set $\partial_n[c] = [a]$. The relation $d_B^2 b = 0$ and injectivity of i show that a is a cycle. Changing the lift by $i(x)$ changes a by $d_A x$, and changing c by a boundary can be accompanied by changing b by a lifted boundary. Thus the formula is independent of both choices. Naturality follows by applying a morphism of short exact sequences to the same lift construction.

Why the missing endpoint zeros really matter. The map $\bar{i} : A_n/B_n(A) \to B_n/B_n(B)$ need not be injective. If $i(a) = d_B b$ for $b \in B_{n+1}$, then a is a cycle and $\pi(b)$ is a cycle in C_{n+1}; the construction gives $\partial_{n+1}[\pi(b)] = [a]$. Conversely every such connecting class maps to zero. Under $H_n(A) \subseteq A_n/B_n(A)$ this identifies $\ker \bar{i}$ with $\operatorname{im} \partial_{n+1}$. Similarly, a cycle in C_{n-1} lifts to a cycle in B_{n-1} precisely when its obstruction ∂_{n-1} vanishes. Hence $\operatorname{coker} \pi_Z \cong \operatorname{im} \partial_{n-1}$, where the latter lies in $H_{n-2}(A)$. The target degree $n-2$ corrects the last index in the handwritten discussion. The manuscript previews a later conceptual explanation: kernels are limit constructions and cokernels are colimit constructions, so they naturally preserve different halves of exactness. The explicit argument above establishes what is needed here without relying on that later language.

Longer exact sequences. For an exact sequence of complexes $0 \to K \to A \xrightarrow{f} B \xrightarrow{g} C \to Q \to 0$, put $I_1 = \operatorname{im} f$ and $I_2 = \operatorname{im} g$. It breaks into $0 \to K \to A \to I_1 \to 0$, $0 \to I_1 \to B \to I_2 \to 0$, and $0 \to I_2 \to C \to Q \to 0$. The successive connecting maps give a composite $H_n(Q) \to H_{n-1}(I_2) \to H_{n-2}(I_1) \to H_{n-3}(K)$. This records the higher connecting construction in the manuscript. It does not by itself assert one ordinary long exact sequence passing through the homology of all five original complexes without the intermediate images.

2.2 The five lemma and two applications

The small photograph on handwritten page 9 contains the diagram redrawn in Figure 6, together with the injective half of the five lemma. Both rows are exact. If b, d are monic and a is epic, then c is monic. To prove this for modules, take $x \in C'$ with $c(x) = 0$. Since $dh'(x) = hc(x) = 0$, injectivity of d gives $h'(x) = 0$, so $x = g'(y)$ for some $y \in B'$. Then $gb(y) = cg'(y) = 0$, hence $b(y) = f(z)$ for some $z \in A$. Lift z through a to $z' \in A'$. Since $b(y - f'(z')) = 0$ and b is injective, $y = f'(z')$, so $x = 0$.

Reversing all arrows proves that c is epic if b, d are epic and e is monic. In particular c is an isomorphism if b, d are isomorphisms, a is epic, and e is monic.

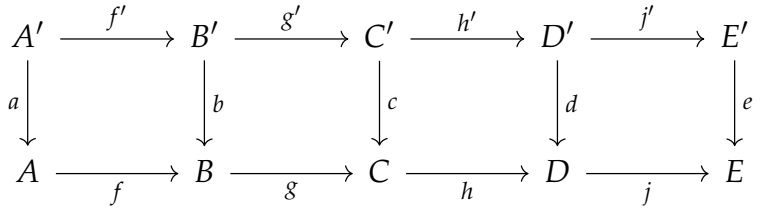


Figure 6. The five lemma diagram recovered from the photograph. Exact rows and all four commutative squares are hypotheses; the conditions on the outer vertical maps control the middle map.

Correction to the cycles-and-boundaries example. Let $Z(C)$ and $B(C)$ denote the complexes with terms $Z_n(C)$ and $B_n(C)$ and zero differentials. We use the manuscript's shift convention $C[r]_n = C_{n+r}$, with differential $(-1)^r d_C$; in particular, $B(C)[-1]_n = B_{n-1}(C)$. There is a short exact sequence $0 \to Z(C) \to C \xrightarrow{d} B(C)[-1] \to 0$. The handwritten example correctly observes that the outer complexes have zero differentials, but then concludes that the connecting map is zero. That conclusion confuses being a boundary in C with being zero in the zero-differential complex $Z(C)$.

The corrected lift computation is direct: for $b \in B_{n-1}(C)$ choose $x \in C_n$ with $dx = b$. The connecting map is $b \mapsto b \in Z_{n-1}(C)$, namely the inclusion $B_{n-1}(C) \hookrightarrow Z_{n-1}(C)$, not the zero map. The long exact sequence consequently recovers $0 \to B_n(C) \to Z_n(C) \to H_n(C) \to 0$. As a check, take $C_1 = C_0 = \mathbb{Z}$ with $d_1 = \mathrm{id}$ and all other terms zero: $H_1(C) = 0$ and $B_0(C) = \mathbb{Z}$, so the handwritten proposed surjection $H_1(C) \to B_0(C)$ cannot exist. This correction preserves the example's purpose, namely computing a connecting map from an explicit short exact sequence.

Acyclic kernel and cokernel. A complex is *acyclic* if every homology group vanishes. The next handwritten example considers a chain map $f : A \to B$ whose kernel K and cokernel Q are acyclic. Put $I = \operatorname{im} f$. The long exact sequences of $0 \to K \to A \to I \to 0$ and $0 \to I \to B \to Q \to 0$ show that $H_n(A) \to H_n(I)$ and $H_n(I) \to H_n(B)$ are isomorphisms for every n. Thus f is a *quasi-isomorphism*, meaning that all induced maps $H_n(f)$ are isomorphisms. This supplies the short completion left implicit at the bottom of page 9.

3 Splittings and chain homotopy

Handwritten pages 10–11 begin with a useful vector-space picture. Over a field choose complements $C_n = Z_n \oplus L_n$ and $Z_n = B_n \oplus H'_n$. The restriction $d_n|_{L_n} : L_n \to B_{n-1}$ is

an isomorphism, while $H'_n \cong H_n(C)$. Thus $C_n \cong B_n \oplus H_n(C) \oplus B_{n-1}$, noncanonically: choices of complements determine the representatives of homology classes.

Define $s_n : C_n \to C_{n+1}$ by projecting onto B_n and then applying $(d_{n+1}|_{L_{n+1}})^{-1}$. For $x = b + h + \ell$ in the chosen decomposition, $ds(x) = b$ and $sd(x) = \ell$. Consequently $s^2 = 0$, ds and sd are projections with zero product, $dsd = d$, and

$$ds + sd = \mathrm{id}_C - \iota_H \pi_H, \tag{3.1}$$

where $\pi_H : C \to H'$ projects onto the zero-differential complex of chosen homology representatives and ι_H is its inclusion. In particular, this chosen s satisfies $ds + sd = \mathrm{id}$ exactly when C is acyclic.

Normalizing a splitting without vector-space assumptions. The manuscript next removes the use of chosen vector-space complements. Suppose C is an exact module complex and a degree-$+1$ map s satisfies $dsd = d$. Set $e = \mathrm{id} - sd$ and $t = se = s(\mathrm{id} - sd)$. Since $de = d - dsd = 0$, $e(x)$ is a cycle and hence a boundary. The identity $dsd = d$ says that ds is the identity on boundaries, so $dse = e$. Moreover $ed = d$, whence $dt = e$, $td = sd$, and therefore $dt + td = \mathrm{id}$. Finally $ese = se - sdse = 0$, so $t^2 = sese = 0$. These identities give an algebraic proof of the handwritten normalization formula without assuming an unexplained decomposition. The identities $(dt)^2 = dt$, $(td)^2 = td$, and $(dt)(td) = (td)(dt) = 0$ then show the complementary projections explicitly.

Split complexes. A complex is *split* if there exists a degree-$+1$ family s with $dsd = d$. It is *split exact* if it is both split and acyclic. The normalization just proved shows that a split exact complex admits t with $dt + td = \mathrm{id}$; such a t is a contracting homotopy. Conversely $dt + td = \mathrm{id}$ gives $dtd = d$ and, for a cycle z, $z = dtz$, so the complex is split exact.

An exact complex that is not split. The manuscript's counterexample is the doubly infinite complex with every term $\mathbb{Z}/4\mathbb{Z}$ and every differential multiplication by 2. Its kernel and image are both $\{0, 2\}$, so it is exact. For any group homomorphism $s : \mathbb{Z}/4\mathbb{Z} \to \mathbb{Z}/4\mathbb{Z}$, however, dsd is multiplication by $4s(1)$ and is zero, whereas d is not zero. Thus no splitting exists. The same example consists of free rank-one modules over $R = \mathbb{Z}/4\mathbb{Z}$, so degreewise freeness alone is insufficient without a boundedness hypothesis.

Exercises retained: Weibel 1.4.1. The source explicitly labels these as exercises still to be done. Show that an acyclic bounded-below complex of free R-modules is split exact. Show also that an acyclic complex of finitely generated free abelian groups is split exact even if it is not bounded below. *Editorial orientation.* In the first assertion,

bounded below means $C_n = 0$ for n sufficiently negative; one begins at the lowest nonzero term and successively splits surjections onto projective boundary modules. In the second, subgroups of free abelian groups are free, so each sequence $0 \to Z_n \to C_n \to Z_{n-1} \to 0$ splits. These are proof routes, rather than a claim that the handwritten exercises already contained completed solutions.

Exercise retained: Weibel 1.4.2. Show that C is split if and only if there are module decompositions $C_n \cong Z_n \oplus B'_n$ and $Z_n \cong B_n \oplus H'_n$, where d_n identifies B'_n with B_{n-1}. Show that it is split exact precisely when the homology summands H'_n vanish. This makes precise which parts of the initial vector-space picture survive over a general ring.

3.1 Homotopies of chain maps

Let $f : C \to D$ be a chain map. It is *null-homotopic* when there are module morphisms $s_n : C_n \to D_{n+1}$ such that $f_n = d^D_{n+1} s_n + s_{n-1} d^C_n$. These maps have degree $+1$ and are not themselves required to be chain maps. Figure 7 recovers the diagonals in the handwritten drawing. A contraction of the identity is a contracting homotopy of the complex; for a general f, we call s a null-homotopy of f.

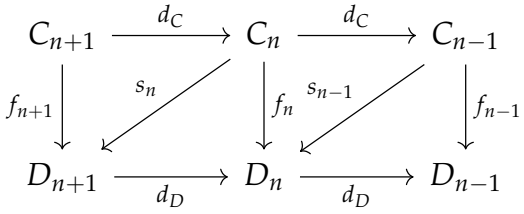


Figure 7. A null-homotopy has diagonals raising the degree by one. The diagram asserts $f_n = d_D s_n + s_{n-1} d_C$, not separate commutativity of either triangle.

Two chain maps $f, g : C \to D$ are *chain homotopic*, written $f \simeq g$, when

$$f - g = d_D s + s d_C. \tag{3.2}$$

For a cycle z, $(f - g)(z) = d_D s(z)$ is a boundary, so $H_n(f) = H_n(g)$. In particular a null-homotopic map induces zero on homology. The proof explains why homotopy is the right equivalence relation for operations that should only depend on homology classes.

Exercise retained and organized: Weibel 1.4.5. The handwritten annotations identify the required steps in constructing the homotopy category K. First show that homotopy is an equivalence relation on the chain maps from C to D: the witnesses are 0, $-s$, and $s + t$ for reflexivity, symmetry, and transitivity. The null-homotopic maps form a subgroup, so the quotient $\mathrm{Hom}_K(C, D)$ is abelian. Next, if $u : B \to C$ and

$v : D \to E$ are chain maps, verify that vsu witnesses $vfu \simeq vgu$. This is the two-sided ideal property needed for composition to be well-defined. Finally, sums of homotopies are homotopies, the zero complex and finite biproducts descend, and K is additive. The quotient functor $Q : \mathsf{Ch} \to K$ is additive, and homology factors as $H_n = \overline{H}_n Q$ because homotopic maps induce the same homology map.

The source ends the exercise by asking whether K is abelian and annotates "No." The appropriate qualification is *not in general*; this is not a universal negative for every coefficient category. For vector spaces over a field the splittings above identify every complex, up to chain homotopy, with its graded homology, and K is equivalent to the abelian category of graded vector spaces. The general-ring obstruction is retained here as the final exercise, rather than replacing an unfinished source question by an unproved blanket assertion.

4 Mapping cones

Handwritten page 12 encodes a chain map $f : B \to C$ into a differential on a larger complex. With the shift convention already fixed, define

$$\operatorname{cone}(f)_n = B_{n-1} \oplus C_n, \qquad d(b, c) = (-d_B b,\, d_C c - f(b)). \tag{4.1}$$

The lower off-diagonal sign is the manuscript's minus sign. The equality $d^2 = 0$ follows because its second component is $-d_C f(b) + f(d_B b) = 0$. For cochain complexes, where the differential raises degree, the corresponding convention is $\operatorname{cone}(f)^n = B^{n+1} \oplus C^n$ with the same formula $d(b, c) = (-d_B b, d_C c - f(b))$. Figure 8 shows the component arrows, so that the diagonal sign can be checked separately from the total differential. Thus the matrix is $\left(\begin{smallmatrix} -d_B & 0 \\ -f & d_C \end{smallmatrix}\right)$ in either grading convention, with the indicated shifted degree of the first summand.

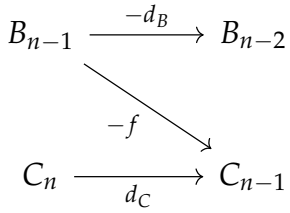


Figure 8. Components of the cone differential. Each column is a direct sum; the downward diagonal is $-f : B_{n-1} \to C_{n-1}$. The missing upward diagonal is the zero map.

Let $i : C \to \operatorname{cone}(f)$ be $c \mapsto (0, c)$. The composite if is null-homotopic: put $s_n(b) = (-b, 0) \in \operatorname{cone}(f)_{n+1}$. Then $ds(b) = (d_B b, f(b))$ and $sd_B(b) = (-d_B b, 0)$, so $ds + sd_B = if$. The inclusion is a chain map, and the composite is zero after passage to the homotopy category, rather than generally zero as a chain map.

Topological motivation. The manuscript compares this construction with attaching a topological cone to a map $X \to Y$, producing $Y \cup_f CX$. The extra shifted summand algebraically records the part that is allowed to become null-homotopic. This is a motivating analogy, not an assertion that arbitrary module complexes are themselves topological spaces.

4.1 The weak cokernel property

Suppose $g : C \to D$ is a chain map and gf is null-homotopic, with $gf = d_D s + s d_B$. Define $\bar{g} : \mathrm{cone}(f) \to D$ by $\bar{g}(b,c) = -s(b) + g(c)$. *Sign correction.* The handwritten formula has $+s(b)$; the minus sign is necessary with the displayed differential $d(b,c) = (-d_B b, d_C c - f(b))$ and the convention $gf = d_D s + s d_B$. It is a chain map: the relation $d_D s + s d_B = gf$ equates $d_D \bar{g}(b,c) = -d_D s(b) + g(d_C c)$ with $\bar{g}\, d(b,c) = s(d_B b) + g(d_C c) - gf(b)$. Moreover $\bar{g} i = g$ strictly. Figure 9 redraws the source's factorization, including the deliberately deleted uniqueness mark.

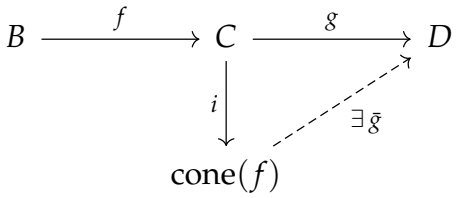


Figure 9. The cone gives a weak cokernel in K: if $[g][f] = 0$, a factorization through $[i]$ exists. The factorization is not asserted to be unique.

Supplementary example. For a concrete failure of uniqueness, let B be a nonzero vector space concentrated in degree 0, and take $C = 0$ and $f = 0$. Then $\mathrm{cone}(f) = B[-1]$ and every endomorphism of $B[-1]$ factors the unique map $0 \to B[-1]$ through $i : 0 \to B[-1]$. The zero and identity endomorphisms are distinct even in K, because the differential is zero. Thus "weak" is essential. The source also previews the dual homotopy fiber $\mathrm{fib}(f) = \mathrm{cone}(f)[1]$. With the present signs it has a chain map $\mathrm{fib}(f) \to B$, $(b,c) \mapsto b$, and gives the dual weak-kernel construction. Its broader role is postponed, as in the manuscript, to the theory of triangulated categories.

Exercise retained with its sign check: Weibel 1.5.1. For $f = \mathrm{id}_C$, the map $s(b,c) = (-c, 0)$ contracts $\mathrm{cone}(\mathrm{id}_C)$. Indeed $ds(b,c) = (d_C c, c)$ and $sd(b,c) = (-d_C c + b, 0)$, whose sum is (b,c). This proves that the identity cone is split exact and verifies the signs in (4.1).

4.2 The cone short exact sequence

There is a degreewise split short exact sequence $0 \to C \xrightarrow{i} \mathrm{cone}(f) \xrightarrow{q} B[-1] \to 0$, where the handwritten quotient map is $q(b,c) = -b$. Since the differential on $B[-1]$

is $-d_B$, this is a chain map. The connecting homomorphism $H_n(B[-1]) \to H_{n-1}(C)$ is $+f_*$ under $H_n(B[-1]) = H_{n-1}(B)$: for a cycle b, lift it to $(-b, 0)$, whose differential is $(0, f(b))$. Consequently the long exact sequence is

$$\cdots \longrightarrow H_n(B) \xrightarrow{f_*} H_n(C) \xrightarrow{i_*} H_n(\mathrm{cone}(f)) \xrightarrow{q_*} H_{n-1}(B) \xrightarrow{f_*} H_{n-1}(C) \longrightarrow \cdots . \tag{4.2}$$

Using the projection $+b$ instead would change the connecting map to $-f_*$; mixing these choices would give an incorrect sign.

The cone criterion. Exactness of (4.2) shows that f is a quasi-isomorphism if and only if $\mathrm{cone}(f)$ is acyclic. If the cone homology vanishes, exactness makes every f_* both injective and surjective. Conversely, if every f_* is an isomorphism, exactness first forces $i_* = 0$ and then forces q_* to be both injective and zero, so every cone homology group vanishes. This completes page 12 and leads to the mapping cylinder on the next page. The relevant background and the numbered exercises above are in [Wei94, Sections 1.2–1.5].

4.3 The mapping cylinder and its relation to the cone

The cylinder separates the map $f : B \to C$ into an inclusion followed by a homotopy equivalence. Retain the cone convention of the preceding subsection, including the minus sign in its off-diagonal entry. Define

$$\mathrm{cyl}(f)_n = B_n \oplus B_{n-1} \oplus C_n, \qquad d(b, b', c) = (d_B b + b', -d_B b', d_C c - f b'). \tag{4.3}$$

Thus the differential has matrix $\begin{pmatrix} d_B & 1 & 0 \\ 0 & -d_B & 0 \\ 0 & -f & d_C \end{pmatrix}$. Its square is zero: the first component is $d_B^2 b + d_B b' - d_B b'$, the second is $d_B^2 b'$, and the third is $d_C^2 c - d_C f b' + f d_B b'$. The chain-map identity $d_C f = f d_B$ cancels the last two terms. Figure 10 records all its nonzero component maps; this is a component diagram, not a claim that paths between different summands commute.

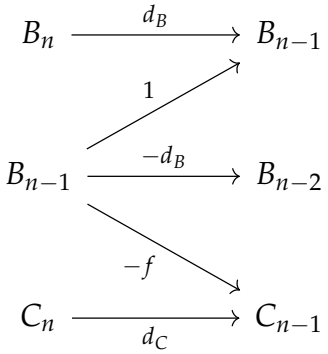


Figure 10. Nonzero entries of the cylinder differential. Each column is a direct sum; the diagonal arrows carry the two off-diagonal entries.

The map $\alpha : C \to \mathrm{cyl}(f)$, $c \mapsto (0,0,c)$, identifies C with a subcomplex. The quotient, initially written in the order (b,b'), has differential $(b,b') \mapsto (d_B b + b', -d_B b')$. Reordering its summands as (b',b) identifies it with $\mathrm{cone}(-\mathrm{id}_B)$. This cone is contractible, so the long exact sequence of $0 \to C \xrightarrow{\alpha} \mathrm{cyl}(f) \to \mathrm{cone}(-\mathrm{id}_B) \to 0$ proves that α is a quasi-isomorphism. This is the argument of the handwritten Lemma 1.5.6.

The stronger conclusion requested in the handwritten Exercise 1.5.4 follows explicitly. Define $\beta : \mathrm{cyl}(f) \to C$ by $\beta(b,b',c) = f(b) + c$. Then $\beta d = f(d_B b + b') + d_C c - f b' = d_C \beta$, and $\beta\alpha = \mathrm{id}_C$. The degree-one map $s(b,b',c) = (0,b,0)$ satisfies

$$(ds + sd)(b,b',c) = (b,b',-f(b)) = (\mathrm{id} - \alpha\beta)(b,b',c).$$

Consequently $\alpha\beta \simeq \mathrm{id}_{\mathrm{cyl}(f)}$, and C and $\mathrm{cyl}(f)$ are isomorphic in the homotopy category. This verifies the source's topological intuition by an actual chain homotopy.

There is also an inclusion $j : B \to \mathrm{cyl}(f)$, $b \mapsto (b,0,0)$, with $\beta j = f$. Quotienting by $j(B)$ gives $\mathrm{cone}(f)$, via $q(b,b',c) = (b',c)$. The two short exact sequences in Figure 11 therefore encode the same map f using the cylinder and the cone. Here $i(c) = (0,c)$ and $p(b',c) = -b'$ are the cone maps already fixed earlier; in particular $q\alpha = i$.

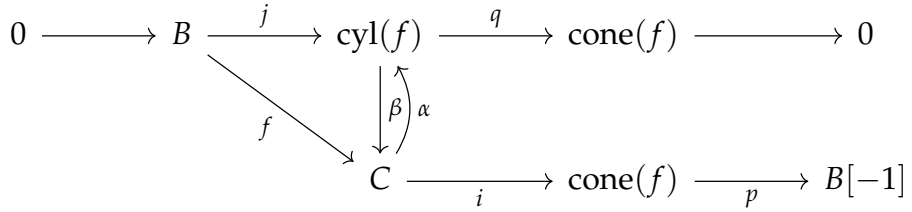


Figure 11. The cylinder sequence is exact, and the lower three terms extend to the exact sequence $0 \to C \xrightarrow{i} \mathrm{cone}(f) \xrightarrow{p} B[-1] \to 0$. The repeated cone denotes the same complex. The relations are $\beta j = f$, $\beta\alpha = \mathrm{id}_C$, and $q\alpha = i$.

Sign check for the small homology diagram. A cone cycle (b',c) has $d_B b' = 0$ and $d_C c = f b'$. Lifting it to $(0,b',c)$ in the cylinder gives differential $(b',0,0)$. Hence the cylinder connecting map is $\partial_{\mathrm{cyl}}[(b',c)] = [b']$, whereas $p_*[(b',c)] = -[b']$. Under the identification by β_*, the cylinder sequence can therefore be written in the exact form

$$\cdots \to H_n(B) \xrightarrow{f_*} H_n(C) \xrightarrow{i_*} H_n(\mathrm{cone}(f)) \xrightarrow{p_*} H_{n-1}(B) \to \cdots,$$

where $p_* = -\partial_{\mathrm{cyl}}$. This explains the minus sign on the connecting arrow in the small pasted diagram on handwritten page 13. One must include this sign when comparing the two presentations; otherwise the apparent square with the last identity map anticommutes.

4.4 When a cone models an ordinary cokernel

Suppose $0 \to B \xrightarrow{f} C \xrightarrow{g} D \to 0$ is a short exact sequence of complexes. In particular f is degreewise monic and D is its cokernel. Define $\varphi : \mathrm{cone}(f) \to D$ by $\varphi(b,c) = g(c)$.

This is a chain map because $gf = 0$, and the diagram in Figure 12 commutes: $\beta j = f$ and $g\beta = \varphi q$. Both rows are short exact.

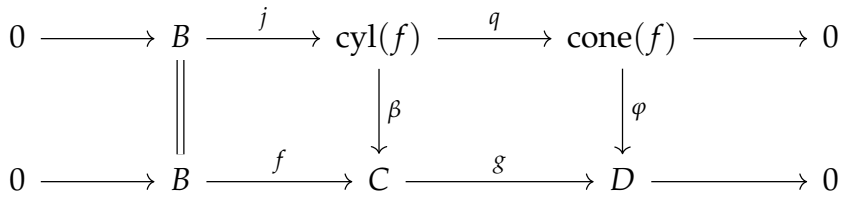


Figure 12. Comparison of the cylinder sequence with an ordinary cokernel sequence. The left vertical map is the identity and the middle map is a homotopy equivalence.

Center the five-term homology ladder at $H_n(\mathrm{cone}(f))$. The surrounding vertical maps are the identity on $H_n(B)$, the isomorphism induced by β in degree n, and the corresponding two maps in degree $n-1$. Four vertical maps are isomorphisms, so the five lemma gives that $H_n(\varphi)$ is an isomorphism for every n. Thus the cone and cokernel agree up to quasi-isomorphism under this monomorphism hypothesis. The source correctly warns that φ need not be a homotopy equivalence.

Supplementary example. Regard $\mathbb{Z} \xrightarrow{\times 2} \mathbb{Z}$ as an injective map of complexes concentrated in degree zero. Its cone is the two-term free complex with differential -2, and $D = \mathbb{Z}/2$ is concentrated in degree zero. The map φ is a quasi-isomorphism. A chain map $D \to \mathrm{cone}(f)$ must be zero in degree zero because $\mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}/2, \mathbb{Z}) = 0$, so no homotopy inverse exists. The distinction is therefore visible in the simplest torsion example.

An additive functor preserves split exact sequences, even when it does not preserve arbitrary exact sequences. To see this, choose maps $i : A \to B$, $p : B \to C$, $r : B \to A$, and $s : C \to B$ with $ri = \mathrm{id}_A$, $ps = \mathrm{id}_C$, $pi = 0$, $rs = 0$, and $ir + sp = \mathrm{id}_B$. Applying an additive functor F preserves all these identities, especially $F(i)F(r) + F(s)F(p) = \mathrm{id}_{F(B)}$. Thus $F(B)$ is the biproduct of $F(A)$ and $F(C)$, and $0 \to F(A) \to F(B) \to F(C) \to 0$ remains split exact. This is the mechanism that will allow derived-functor constructions to use degreewise split resolutions.

5 Functor categories, representable functors, and exactness

This section follows handwritten pages 15–18. A category is *small* when its objects and morphisms form sets; it is *locally small* when each individual collection $\mathrm{Hom}(A, B)$ is a set. The category of sets is locally small but not small in the usual foundational convention. The source's reference to Russell's paradox concerns precisely the distinction between a set and a proper class: one cannot treat the collection of all sets as an ordinary set. A universe convention, or the explicit smallness assumptions below, keeps the functor constructions well defined.

A functor $E : \mathcal{A} \to \mathcal{B}$ is *faithful* if every map $\mathrm{Hom}_{\mathcal{A}}(A, A') \to \mathrm{Hom}_{\mathcal{B}}(EA, EA')$ is injective, and *full* if each is surjective. Thus a fully faithful functor identifies the morphisms between its objects without losing or adding any. For abelian categories, an additive functor is *exact* if it preserves short exact sequences. Saying that a subcategory is full refers to its inclusion functor, rather than to a new structure on its objects.

The *Freyd–Mitchell embedding theorem* states that a small abelian category admits a fully faithful exact functor into a category of modules over some ring. The module category itself is generally large but locally small. This theorem explains why element arguments often model finite abelian-category calculations. The handwritten source labels its proof as unfinished; the outline below preserves that status rather than treating its last proposed step as established.

Closure inside an abelian category. Let $\mathcal{C}$ be a full subcategory of an abelian category $\mathcal{A}$. All closure statements here are understood up to isomorphism. The inherited addition makes $\mathcal{C}$ additive precisely when it contains a zero object and the ambient finite biproducts of its objects. Indeed, biproduct identities internal to $\mathcal{C}$ are the same identities in $\mathcal{A}$, while ambient biproducts lying in $\mathcal{C}$ have their universal properties there by fullness. Moreover, $\mathcal{C}$ is abelian and its inclusion is exact precisely when it is additive and closed under ambient kernels and cokernels of its morphisms. In the latter direction, the ambient coimage-to-image isomorphism supplies the abelian axiom inside $\mathcal{C}$; in the former direction, exactness of the inclusion preserves those kernels and cokernels. The adjective *ambient* matters: an intrinsically abelian full subcategory need not have an exact inclusion. The source points toward sheaves inside presheaves as a later test of this distinction; a full development of sheafification is not needed here.

Let $\mathcal{D}$ be small and $\mathcal{A}$ abelian. The functor category $\mathcal{A}^{\mathcal{D}}$ has functors $F : \mathcal{D} \to \mathcal{A}$ as objects and natural transformations as morphisms. A natural transformation $\eta : F \Rightarrow G$ is a family $\eta_d : F(d) \to G(d)$ satisfying $\eta_{d'} F(u) = G(u)\eta_d$ for every $u : d \to d'$, as in Figure 13. Kernels, cokernels, and exactness in this category are computed object by object.

$$\begin{array}{ccccc} K(d) & \xrightarrow{k_d} & F(d) & \xrightarrow{\eta_d} & G(d) \\ {\scriptstyle K(u)}\big\downarrow & & \big\downarrow{\scriptstyle F(u)} & & \big\downarrow{\scriptstyle G(u)} \\ K(d') & \xrightarrow[k_{d'}]{} & F(d') & \xrightarrow[\eta_{d'}]{} & G(d') \end{array}$$

Figure 13. Naturality induces the map on pointwise kernels. The right square commutes by definition, and the left one determines $K(u)$ uniquely.

For the proof, put $K(d) = \ker \eta_d$. Naturality gives $\eta_{d'} F(u) k_d = G(u)\eta_d k_d = 0$, so the universal property of $k_{d'}$ gives a unique $K(u)$ with $k_{d'} K(u) = F(u) k_d$. Uniqueness

proves $K(\mathrm{id}) = \mathrm{id}$ and $K(vu) = K(v)K(u)$. The same property gives the universal factorization of any natural map annihilated by η. Cokernels are constructed dually, and the coimage-to-image comparison is an isomorphism at every object, hence a natural isomorphism. This proves that $\mathcal{A}^{\mathcal{D}}$ is abelian. The structure maps are forced by universal properties; they are not additional arbitrary data.

If $\mathcal{D}$ is the discrete category on the integers, $\mathrm{Ab}^{\mathcal{D}}$ is exactly the category of graded abelian groups. *Editorial clarification:* a discrete category has identity arrows and no other arrows, rather than an empty collection of morphisms. If $\mathcal{D}$ is the ordered set of integers, with an arrow $n \to m$ for $n \leq m$, a functor gives a sequence of maps. The full subcategory in which successive maps have zero composite is the category of cochain complexes; this extra relation is essential. Finally, for the poset $\mathcal{O}(X)$ of open subsets of a space, presheaves valued in $\mathcal{A}$ are contravariant functors, so $\mathrm{PSh}(X;\mathcal{A}) = \mathcal{A}^{\mathcal{O}(X)^{\mathrm{op}}}$ is abelian.

5.1 Left exactness of Hom and detection of exactness

For an additive covariant functor $F : \mathcal{A} \to \mathcal{B}$, *left exact* means that every short exact sequence $0 \to A \to B \to C \to 0$ induces an exact sequence $0 \to F(A) \to F(B) \to F(C)$; *right exact* means that $F(A) \to F(B) \to F(C) \to 0$ is exact. A contravariant functor is called left exact when the corresponding covariant functor $\mathcal{A}^{\mathrm{op}} \to \mathcal{B}$ is left exact. It consequently sends the same short exact sequence to $0 \to F(C) \to F(B) \to F(A)$.

Fix $M \in \mathcal{A}$. The functor $\mathrm{Hom}_{\mathcal{A}}(M, -)$ is left exact. For $0 \to A \xrightarrow{f} B \xrightarrow{g} C \to 0$, the induced maps are postcomposition. If $f\alpha = 0$, then $\alpha = 0$ because f is monic. If $g\beta = 0$ for $\beta : M \to B$, the kernel property of f gives a unique $\alpha : M \to A$ with $\beta = f\alpha$. These are exactly injectivity of f_* and $\ker g_* = \mathrm{im}\, f_*$. The correct dual assertion is left exactness of the *contravariant* functor $\mathrm{Hom}_{\mathcal{A}}(-, M) = \mathrm{Hom}_{\mathcal{A}^{\mathrm{op}}}(M, -)$; it does not claim that $\mathrm{Hom}(M, -)$ preserves epimorphisms.

The representable functor $h_A = \mathrm{Hom}_{\mathcal{A}}(-, A)$ acts on $u : M \to N$ by precomposition, $h_A(u) : \mathrm{Hom}(N, A) \to \mathrm{Hom}(M, A)$, $a \mapsto au$. A map $v : A \to B$ determines a natural transformation $h_A \to h_B$ by postcomposition. Conversely, for a natural transformation $\eta : h_A \to h_B$, set $v = \eta_A(\mathrm{id}_A)$. Naturality with respect to $a : M \to A$ implies $\eta_M(a) = va$, so this construction is bijective. This is the representable case of the Yoneda lemma, and proves the fully faithful Yoneda embedding $A \mapsto h_A$.

The manuscript next uses a reflection principle. If for every M the sequence

$$\mathrm{Hom}(M, A) \xrightarrow{\alpha_*} \mathrm{Hom}(M, B) \xrightarrow{\beta_*} \mathrm{Hom}(M, C)$$

is exact at its middle term, then $A \xrightarrow{\alpha} B \xrightarrow{\beta} C$ is exact at B. First test $M = A$ on id_A to obtain $\beta\alpha = 0$. Next let $k : K \to B$ be $\ker \beta$ and test $M = K$. Since $\beta k = 0$, exactness gives $\sigma : K \to A$ with $\alpha\sigma = k$. Thus $\ker \beta \subseteq \mathrm{im}\, \alpha$; the reverse inclusion

follows from $\beta\alpha = 0$. *Editorial clarification:* this is a sufficient test, not an assertion that Hom preserves every sequence exact only at its middle term. For example, $\mathbb{Z} \twoheadrightarrow \mathbb{Z}/2 \to 0$ is exact, but applying $\mathrm{Hom}(\mathbb{Z}/2, -)$ is not exact at the middle group. For a sequence starting with $0 \to A$, left exactness and this reflection principle give the familiar equivalence of left-exactness tests.

5.2 The unfinished embedding argument

Assume now that $\mathcal{A}$ is small and abelian. Write $\mathcal{F} = \mathrm{Add}(\mathcal{A}^{\mathrm{op}}, \mathrm{Ab})$ for the category of additive contravariant functors, and $\mathcal{L}$ for its full subcategory of left exact functors. The Yoneda embedding lands in $\mathcal{L}$. To measure its failure to be right exact as a functor into $\mathcal{F}$, start with $0 \to A \xrightarrow{f} B \xrightarrow{g} C \to 0$ and define

$$W(M) = \mathrm{coker}\big(\mathrm{Hom}(M, B) \xrightarrow{g_*} \mathrm{Hom}(M, C)\big).$$

These cokernels form a contravariant additive functor, and there is a pointwise exact sequence $0 \to h_A \to h_B \to h_C \to W \to 0$.

A contravariant functor W is *weakly effaceable* if for every M and every $x \in W(M)$ there is an epimorphism $p : P \twoheadrightarrow M$ with $W(p)(x) = 0$. Here the epimorphism may depend on the element.

Editorial bridge: pullbacks. For maps $a : M \to C$ and $g : B \to C$, their pullback is an object $P = M \times_C B$ with projections $p : P \to M$ and $q : P \to B$ satisfying $ap = gq$. It is universal among such compatible pairs: maps $u : X \to M$ and $v : X \to B$ with $au = gv$ determine a unique map $X \to P$ with components u, v. In modules, $P = \{(m, b) : a(m) = g(b)\}$; in any abelian category, it is the kernel of $(a, -g) : M \oplus B \to C$.

If g is epic, then p is epic. For modules, lift $a(m)$ through g to obtain a compatible pair over each m. The categorical proof uses the same kernel construction. Put $t = (a, -g)$; it is epic because its restriction to B is $-g$. If $h : M \to X$ satisfies $hp = 0$, then $(h, 0)$ annihilates $\ker t$ and factors through t as $wt = (h, 0)$, since an epimorphism in an abelian category is the cokernel of its kernel. Restricting to B gives $wg = 0$, hence $w = 0$ and then $h = 0$. This is the cancellation test for p to be epic.

The particular defect functor above is weakly effaceable: represent x by $a : M \to C$ and form this pullback using the epimorphism $g : B \to C$. The epic projection p satisfies $ap = gq$, so ap represents zero in $W(P)$. This supplies the omitted mechanism behind the source's assertion.

The pasted argument on handwritten page 18 proves that every natural transformation $\tau : W \to L$, with W weakly effaceable and $L \in \mathcal{L}$, is zero. Choose p killing x; Figure 14 gives $L(p)\tau_M(x) = \tau_P W(p)(x) = 0$. If $K = \ker p$, left exactness sends $0 \to K \to P \to M \to 0$ to $0 \to L(M) \xrightarrow{L(p)} L(P) \to L(K)$. Hence $L(p)$ is injective and $\tau_M(x) = 0$. This holds for every M and x.

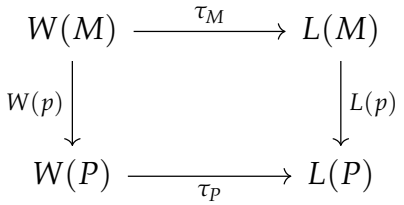


Figure 14. Naturality in the effacement argument. The chosen element vanishes on the left after pullback, while the right vertical map is injective.

The manuscript proposes an exact reflector $\mathcal{R} : \mathcal{F} \to \mathcal{L}$ left adjoint to the inclusion $i : \mathcal{L} \hookrightarrow \mathcal{F}$, with the weakly effaceable functors as its kernel. It also notes that a localization theorem is needed to justify this construction. Conditional on that adjunction, the preceding calculation gives $\mathrm{Hom}_{\mathcal{L}}(\mathcal{R}W, L) \cong \mathrm{Hom}_{\mathcal{F}}(W, iL) = 0$. Taking $L = \mathcal{R}W$ shows $\mathrm{id}_{\mathcal{R}W} = 0$, hence $\mathcal{R}W = 0$. If the stated exactness and reflector properties have been established, applying $\mathcal{R}$ to the defect sequence makes $0 \to h_A \to h_B \to h_C \to 0$ exact in $\mathcal{L}$, even though the last map need not be an epimorphism in the larger functor category. This is the source's useful idea that changing the ambient category changes the relevant cokernel.

Status of the source outline. The last handwritten step proposes a faithful projective object $P \in \mathcal{L}$ and the ring $R = \mathrm{End}_{\mathcal{L}}(P)$. The right action on $\mathrm{Hom}_{\mathcal{L}}(P, A)$ would be $a \cdot r = a \circ r$, using composition as multiplication in R. A projective P would make this Hom functor exact, and a generator would make it faithful. Its existence and the further hypotheses needed for full faithfulness, however, are not proved in the notes, and cannot be inferred from the construction of $\mathcal{L}$ alone. In particular, left-exact-functor categories need not supply the proposed projective generator. We therefore retain this as an incomplete proposed route, together with the source's explicit instruction that the details remain to be considered. The Freyd–Mitchell theorem stated above is not being derived from this unproved last step. The published errata also correct the relevant statement on page 29 of Weibel: the conclusion concerns every *small full abelian subcategory* of $\mathcal{L}$, rather than an equivalence of the entire $\mathcal{L}$ with a module category [Wei94, Section 1.6][Wei].

6 Delta functors and projective resolutions

The next part follows handwritten pages 19–23. A homological δ-functor from an abelian category $\mathcal{A}$ to an abelian category $\mathcal{B}$ consists of additive functors $T_n : \mathcal{A} \to \mathcal{B}$ for $n \geq 0$, together with connecting maps $\delta_n : T_n(C) \to T_{n-1}(A)$ for every short exact sequence $0 \to A \to B \to C \to 0$. Set $T_n = 0$ for $n < 0$. The requirements are exactness of the associated long sequence and naturality of its connecting maps with respect to morphisms of short exact sequences. Thus the sequence ends in $T_0(A) \to T_0(B) \to$

$T_0(C) \to 0$, and T_0 is right exact. Dually, a cohomological δ-functor T^n has connecting maps $T^n(C) \to T^{n+1}(A)$ and starts with $0 \to T^0(A) \to T^0(B) \to T^0(C)$, making T^0 left exact. Figure 15 records the homological compatibility at the boundary map; the other squares arise from functoriality.

$$\begin{array}{ccccccc} T_n(B') & \longrightarrow & T_n(C') & \xrightarrow{\delta'_n} & T_{n-1}(A') & \longrightarrow & T_{n-1}(B') \\ \downarrow & & \downarrow & & \downarrow & & \downarrow \\ T_n(B) & \longrightarrow & T_n(C) & \xrightarrow[\delta_n]{} & T_{n-1}(A) & \longrightarrow & T_{n-1}(B) \end{array}$$

Figure 15. A morphism from $0 \to A' \to B' \to C' \to 0$ to $0 \to A \to B \to C \to 0$ induces a commuting ladder. Naturality of δ is an additional part of a delta functor's data.

6.1 Torsion and why the zeroth functor is not enough

Fix an integer p. Multiplication by p is a natural endomorphism of the identity functor on abelian groups. Apply the snake lemma to the short exact sequence $0 \to A \to B \to C \to 0$ and its multiplication-by-p copy. With $A[p] = \{a \in A : pa = 0\}$, the resulting exact sequence is

$$0 \to A[p] \to B[p] \to C[p] \xrightarrow{\delta} A/pA \to B/pB \to C/pC \to 0.$$

Thus $S_0(A) = A/pA$, $S_1(A) = A[p]$, and $S_n(A) = 0$ for $n \geq 2$ form a homological δ-functor. To see its boundary directly, lift $c \in C[p]$ to $b \in B$; then pb comes from a unique $a \in A$, and $\delta(c) = [a] \in A/pA$. A different lift changes a by a multiple of p, exactly as required.

The source generalizes this construction to a unital ring R and $r \in R$, working from left R-modules to abelian groups. Set $M[r] = \{m : rm = 0\}$ and define $S_0(M) = M/rM$, $S_1(M) = M[r]$, and $S_n = 0$ for $n \geq 2$. Multiplication by r commutes with every R-linear map, so the same snake-lemma construction is natural as a construction of abelian groups. No centrality of r is needed for this target category. If the values are instead required to be R-modules with these maps R-linear, an additional hypothesis such as centrality is needed. The quotient R/rR is a right R-module.

Editorial reminder: the tensor product. If N is a right R-module and M a left R-module, $N \otimes_R M$ is the abelian group generated by symbols $n \otimes m$, subject to additivity in each variable and the balancing relation $(nr) \otimes m = n \otimes (rm)$. Thus additive balanced maps out of pairs (n, m) factor uniquely through this group. The basic evaluation maps are $R \otimes_R M \cong M$, $r \otimes m \mapsto rm$, and $N \otimes_R R \cong N$, $n \otimes r \mapsto nr$; their inverses insert 1 in the other factor. A homomorphism $d : N \to N'$ induces $(d \otimes 1)(n \otimes m) = d(n) \otimes m$.

In particular, $(R/rR) \otimes_R M \cong M/rM$ via $[a] \otimes m \mapsto am + rM$, with inverse $m + rM \mapsto [1] \otimes m$. The balancing relation verifies both formulas and kills exactly the

added relation $rm = 0$. These are the rules used in the following calculation; tensoring does not generally preserve an exact sequence.

To compare this with the other construction mentioned in the source, recall the intended meaning of $\mathrm{Tor}_n^R(N, M)$: choose a projective resolution $P_\bullet \to N$ of the right R-module N, tensor with the left R-module M, and take degree-n homology. The existence and independence of resolutions are established below and later in these notes; the following explicit free resolution makes the comparison computable without assuming those theorems. The functor $T_n(M) = \mathrm{Tor}_n^R(R/rR, M)$ has the same degree-zero functor as S_n, but its positive-degree terms can differ.

The concrete example on page 19 is $R = \mathbb{Z}/4$ and $r = 2$. The right R-module $R/2R$ has the free resolution

$$\cdots \xrightarrow{\times 2} R \xrightarrow{\times 2} R \xrightarrow{\times 2} R \longrightarrow R/2R \longrightarrow 0,$$

because the kernel and image of multiplication by 2 are both $2R = \{0, 2\}$. Tensoring with $M = R$ leaves this exact resolution unchanged in positive degrees, so $T_1(R) = 0$, whereas $S_1(R) = R[2] = 2R \neq 0$. Tensoring instead with $M = R/2R$ identifies each term with M and sends m to $2m = 0$ under every differential, hence $T_n(R/2R) \cong R/2R$ for every $n \geq 0$. In contrast, S_n vanishes for all $n \geq 2$.

The annihilator correction visible in the source can also be made explicit without a long-exact-sequence argument. Let $K_r = \{a \in R : ra = 0\}$, a right ideal. Begin a free right-module resolution of R/rR with $P_0 = R$, $P_1 = R$, and $d_1(a) = ra$. Choose a free module P_2 mapping onto K_r, and continue by resolving successive kernels. After tensoring with M, the kernel of $d_1 \otimes 1$ is $M[r]$, while the image of $d_2 \otimes 1$ is K_rM, the subgroup of finite sums am with $a \in K_r$. Hence

$$\mathrm{Tor}_1^R(R/rR, M) \cong M[r]/K_rM.$$

Here $K_rM \subseteq M[r]$ since $r(am) = (ra)m = 0$. If $K_r = 0$, the two-term free resolution $0 \to R \xrightarrow{r\cdot} R \to R/rR \to 0$ shows that $T_1(M) = M[r]$ and $T_n(M) = 0$ for $n \geq 2$. If $K_r \neq 0$, taking $M = R$ already distinguishes S_1 and T_1. The lesson of the example is the source's own motivation: a prescribed degree-zero functor alone does not determine the higher terms; a universal property is needed.

6.2 Morphisms and universality

A morphism of homological δ-functors $\phi : S_* \to T_*$ is a family of natural transformations $\phi_n : S_n \to T_n$ satisfying $\delta_n^T \phi_n^C = \phi_{n-1}^A \delta_n^S$ for every short exact sequence. A homological δ-functor T_* is *universal* if, for every other S_*, every natural transformation $\phi_0 : S_0 \to T_0$ extends uniquely to such a morphism $S_* \to T_*$. Thus the extension points *toward* the universal homological functor. In the cohomological convention the

arrows reverse: universality of T^* means that every $\phi^0 : T^0 \to S^0$ extends uniquely to $T^* \to S^*$. These are the left and right satellite constructions, respectively, in the terminology recorded in the manuscript; left and right derived functors will provide the relevant universal examples.

If F is exact, the family with degree-zero term F and all higher terms zero is universal in both senses. The higher components of a morphism must be zero. To justify boundary compatibility, which the brief source proof leaves implicit, consider the homological case at degree one. Naturality gives $F(i)\phi_0^A\delta^S = \phi_0^B S_0(i)\delta^S = 0$, where $i : A \to B$. Since $F(i)$ is monic, $\phi_0^A\delta^S = 0$, as needed. The cohomological argument is dual: use the epimorphism $F(B) \to F(C)$ to force the corresponding boundary composite to vanish. There are no other nontrivial boundary conditions.

If two universal homological δ-functors have an identified degree-zero functor, universality gives maps between them extending the identity in both directions. Their composites must be identities by uniqueness. They are therefore uniquely isomorphic by an isomorphism compatible with the chosen degree-zero identification. The same argument applies cohomologically. This conclusion is more precise than literal equality of all chosen constructions.

6.3 Projective objects and lifting

An object P of an abelian category is *projective* if, for every epimorphism $g : B \twoheadrightarrow C$ and every $\gamma : P \to C$, there exists a morphism $\beta : P \to B$ with $g\beta = \gamma$. Figure 16 shows both the lifting problem and the pullback used in its splitting criterion. The dotted lift need not be unique: the difference of two lifts factors through $\ker g$. In a module category, merely choosing an inverse image of each element does not suffice, because those choices must assemble into a linear map.

Since $\mathrm{Hom}(P, -)$ is already left exact, projectivity is equivalent to exactness of $\mathrm{Hom}(P, -)$: its remaining surjectivity condition is exactly the lifting property. An existing coproduct $\coprod_{i\in I} P_i$ of projectives is projective as well. Lift the restriction of γ on every summand and combine those lifts using the coproduct property. For an infinite family this uses the usual set-theoretic choice of lifts, as in the standard module conventions.

An object P is projective if and only if every epimorphism $q : E \twoheadrightarrow P$ splits. Projectivity applied to id_P produces a section $s : P \to E$ with $qs = \mathrm{id}_P$. If $i : K \to E$ is the kernel of q, define $r : E \to K$ by $ir = \mathrm{id}_E - sq$. Then $ri = \mathrm{id}_K$, $rs = 0$, and $ir + sq = \mathrm{id}_E$, so $E \cong K \oplus P$. For modules this is the source's element decomposition $x = (x - sq(x)) + sq(x)$, whose first summand lies in K and whose second lies in $s(P)$; their intersection is zero.

Conversely, suppose all epimorphisms onto P split. Form the pullback of g along γ. In modules it is $E = \{(p, b) \in P \oplus B : \gamma(p) = g(b)\}$, and $\pi_P : E \to P$ is surjective. Cat-

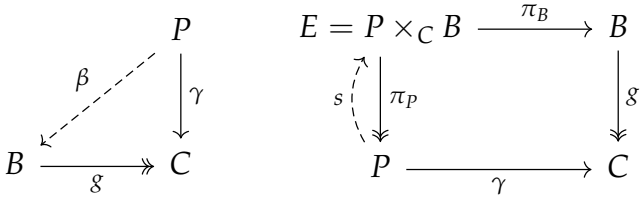


Figure 16. Projectivity solves the left lifting problem. On the right, a section of the pullback epimorphism gives the desired lift $\beta = \pi_B s$.

egorically this uses stability of epimorphisms under pullback in an abelian category. A section s gives $\beta = \pi_B s$, and the pullback identity gives $g\beta = \gamma\pi_P s = \gamma$. This proves the converse. It also supplies a practical failure test: finding one nonsplit epimorphism onto P proves that P is not projective.

For $n \geq 2$, $\mathbb{Z} \twoheadrightarrow \mathbb{Z}/n$ cannot split: a section would send the nonzero torsion generator to an element of the torsion-free group $\mathbb{Z}$. Hence $\mathbb{Z}/n$ is not projective over $\mathbb{Z}$, although it is free of rank one, and therefore projective, over $\mathbb{Z}/n$. The coefficient ring is part of the assertion.

6.4 Free modules and direct summands

A free left R-module with basis $(e_i)_{i\in I}$ is $F = \bigoplus_{i\in I} Re_i$: every element has a unique expression as a finite sum $\sum_i r_i e_i$. Equivalently, any choice of elements m_i in a left R-module M determines a unique linear map $F \to M$ by $\sum_i r_i e_i \mapsto \sum_i r_i m_i$. Finite support is essential when I is infinite. The module $\mathbb{Z}/n$ is finitely generated over $\mathbb{Z}$ but not free there, because $n\overline{1} = 0$ is a nontrivial relation over $\mathbb{Z}$; it is free over $\mathbb{Z}/n$, where the coefficient n is already zero.

Every free module is projective. For a surjection $g : B \to C$ and a map $\gamma : F \to C$, choose $b_i \in B$ lifting $\gamma(e_i)$ and extend $e_i \mapsto b_i$ linearly. Every module M is also a quotient of a free module: choose generators (m_i), or simply use every element of its underlying set, and map the corresponding formal basis vectors to those elements. Thus $M \cong F/\ker(F \twoheadrightarrow M)$.

A module P is projective if and only if it is a direct summand of a free module. If P is projective, the canonical surjection $\mathrm{Fr}(P) = \bigoplus_{p\in P} R[p] \to P$, $[p] \mapsto p$, splits, yielding $\mathrm{Fr}(P) \cong P \oplus \ker\pi$. Conversely, suppose $F \cong P \oplus Q$ is free, with inclusion $i : P \to F$ and projection $\pi : F \to P$. Given $g : B \twoheadrightarrow C$ and $\gamma : P \to C$, lift $\gamma\pi : F \to C$ to $a : F \to B$. Then $\beta = ai$ satisfies $g\beta = \gamma\pi i = \gamma$. As emphasized in the handwriting, being a direct summand is stronger than merely being a submodule; the retraction is the crucial extra datum.

The category of finite abelian groups provides a contrasting example: it has no nonzero projective objects. If $P \neq 0$, the structure theorem gives $P \cong \mathbb{Z}/p^m \oplus Q$ for some prime p and $m \geq 1$. Consider the surjection $g : \mathbb{Z}/p^{m+1} \oplus Q \to \mathbb{Z}/p^m \oplus Q$

that reduces the first coordinate. If it had a section, then $s(\overline{1},0) = (\overline{a},0)$ with $a \equiv 1 \pmod{p^m}$. But $p^m(\overline{1},0) = 0$ would imply $p^m\overline{a} = 0$ modulo p^{m+1}, whereas $a \equiv 1 \pmod{p^m}$ gives $p^m a \equiv p^m \neq 0$ modulo p^{m+1}. This contradiction proves the claim. The epimorphism used here remains inside the finite category.

6.5 Existence and comparison of projective resolutions

An abelian category has *enough projectives* if each object is the target of an epimorphism from a projective object. A left resolution of M is a nonnegative chain complex $P_\bullet$, together with an augmentation $\epsilon : P_0 \to M$, such that $\cdots \to P_2 \to P_1 \to P_0 \xrightarrow{\epsilon} M \to 0$ is exact. It is a projective resolution when all P_n are projective.

If there are enough projectives, choose $P_0 \twoheadrightarrow M$ and let K_0 be its kernel. Choose $P_1 \twoheadrightarrow K_0$, let K_1 be its kernel, and continue. The differential $d_n : P_n \to P_{n-1}$ is the composite of $P_n \twoheadrightarrow K_{n-1}$ with $K_{n-1} \hookrightarrow P_{n-1}$. These composites have square zero and image equal to the next kernel by construction. Figure 17 redraws the kernel factorization in the handwritten figure. In particular, every module admits a free, hence projective, resolution.

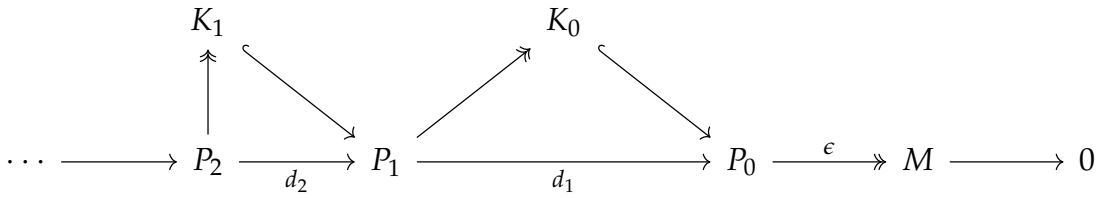


Figure 17. A resolution constructed by successive kernels. Each differential factors through the preceding kernel; the short exact pieces are $0 \to K_n \to P_n \to K_{n-1} \to 0$, with $K_{-1} = M$.

Comparison theorem. Let $P_\bullet \xrightarrow{\epsilon} M$ be an augmented nonnegative complex, so $\epsilon d_1^P = 0$, with projective P_n, and let $Q_\bullet \xrightarrow{\eta} N$ be an exact resolution. For every $f' : M \to N$ there is a chain map $f : P_\bullet \to Q_\bullet$ satisfying $\eta f_0 = f'\epsilon$, and any two such lifts are chain homotopic. This includes the usual case where P is a projective resolution. The handwritten correction is useful: exactness of the source complex is not required, and projectivity of the target terms is not required. The proof first constructs a lift degree by degree, then applies the same lifting mechanism to the difference of two lifts.

At degree zero, projectivity of P_0 lifts $f'\epsilon : P_0 \to N$ through $\eta : Q_0 \twoheadrightarrow N$. Suppose $f_0, \dots, f_n$ have been chosen with the required commutation identities. Put $\gamma_{n+1} = f_n d_{n+1}^P$. If $n \geq 1$, then $d_n^Q \gamma_{n+1} = f_{n-1} d_n^P d_{n+1}^P = 0$; if $n = 0$, then $\eta\gamma_1 = f'\epsilon d_1^P = 0$. Set $Z_0(Q) = \ker \eta$ and $Z_n(Q) = \ker d_n^Q$ for $n \geq 1$. Thus γ_{n+1} factors through $Z_n(Q)$. Exactness of the augmented target makes $Q_{n+1} \twoheadrightarrow Z_n(Q)$ epic, so projectivity of P_{n+1} provides a lift satisfying

$$d_{n+1}^Q f_{n+1} = f_n d_{n+1}^P. \tag{6.1}$$

Figure 18 distinguishes the differential, which need not be epic onto Q_n, from its epic factor onto cycles. This distinction is the central point of the handwritten induction.

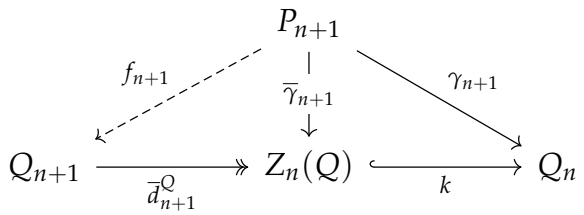


Figure 18. The comparison lift factors through cycles. The right triangle is the factorization $k\overline{\gamma}_{n+1} = f_n d^P_{n+1}$, and projectivity supplies the left triangle. Their composite is $d^Q_{n+1} f_{n+1} = f_n d^P_{n+1}$.

This proves existence. The next paragraph proves uniqueness up to chain homotopy and extracts independence of the chosen projective resolution.

Uniqueness of the comparison lift. We continue the proof of the comparison theorem in Section 6.5, using its augmentations $\epsilon : P_0 \to M$ and $\eta : Q_0 \to N$. If $f, g : P_\bullet \to Q_\bullet$ lift the same map $f' : M \to N$, put $h = f - g$. We construct maps $s_n : P_n \to Q_{n+1}$ satisfying $h_n = d^Q_{n+1} s_n + s_{n-1} d^P_n$. Set $s_{-1} = 0$. In degree zero, $\eta h_0 = (f' - f')\epsilon = 0$, so h_0 factors through $\ker \eta$. Exactness of the augmented target gives an epimorphism $Q_1 \to \ker \eta$, and projectivity of P_0 lifts this factorization to s_0. Thus $h_0 = d^Q_1 s_0$.

Suppose $s_0, \dots, s_{n-1}$ have been constructed. Define $r_n = h_n - s_{n-1} d^P_n$. The chain-map identity and the preceding homotopy identity give

$$d^Q_n r_n = d^Q_n h_n - (h_{n-1} - s_{n-2} d^P_{n-1}) d^P_n = h_{n-1} d^P_n - h_{n-1} d^P_n = 0.$$

Consequently r_n factors through $Z_n(Q) = \ker d^Q_n$. Exactness supplies $Q_{n+1} \twoheadrightarrow Z_n(Q)$, and projectivity again produces s_n with $d^Q_{n+1} s_n = r_n$. Figure 19 displays precisely the lifting step; its right-hand vertical map is the factorization of r_n, not a map defined using an as-yet unknown s_n.

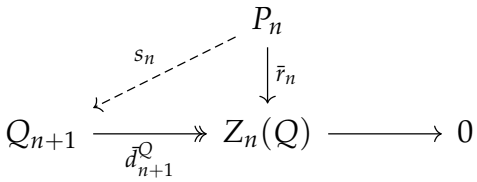


Figure 19. The induction proving uniqueness of comparison maps up to chain homotopy. The lower row is exact.

This proves $f \simeq g$ without requiring the source complex to be exact or the target terms to be projective.

Uniqueness of projective resolutions. If $P_\bullet \to M$ and $Q_\bullet \to M$ are both projective resolutions, comparison gives maps $f : P_\bullet \to Q_\bullet$ and $g : Q_\bullet \to P_\bullet$ lifting id_M. Both gf and id_P lift id_M, hence $gf \simeq \mathrm{id}_P$; similarly $fg \simeq \mathrm{id}_Q$. The resolutions are therefore chain homotopy equivalent. This conclusion is an equivalence up to homotopy, not an isomorphism of complexes term by term.

6.6 The horseshoe construction

The handwritten construction next starts from a short exact sequence $0 \to A' \xrightarrow{i} A \xrightarrow{\pi} A'' \to 0$ and projective resolutions $P'_\bullet \to A'$ and $P''_\bullet \to A''$. The *horseshoe lemma* constructs a projective resolution $P_\bullet \to A$ with $P_n = P'_n \oplus P''_n$ and a short exact sequence of complexes $0 \to P'_\bullet \to P_\bullet \to P''_\bullet \to 0$ that is split in each degree. The construction chooses splittings of the underlying graded objects; it does not assert a splitting by chain maps.

First lift $\epsilon'' : P''_0 \to A''$ along π to $\beta : P''_0 \to A$. Define $\epsilon : P'_0 \oplus P''_0 \to A$ by $\epsilon = i\epsilon' + \beta$, meaning $\epsilon(x', x'') = i\epsilon'(x') + \beta(x'')$ in a module category. The maps in Figure 20 commute because $\pi\beta = \epsilon''$ and $\pi i = 0$.

$$\begin{array}{ccccccccc}
0 & \longrightarrow & P'_0 & \hookrightarrow & P'_0 \oplus P''_0 & \twoheadrightarrow & P''_0 & \longrightarrow & 0 \\
 & & \downarrow{\scriptstyle \epsilon'} & & \downarrow{\scriptstyle \epsilon} & & \downarrow{\scriptstyle \epsilon''} & & \\
0 & \longrightarrow & A' & \xrightarrow[i]{} & A & \xrightarrow[\pi]{} & A'' & \longrightarrow & 0
\end{array}$$

Figure 20. The degree-zero horseshoe. The upper row is the standard biproduct sequence and both rows are exact.

The snake lemma gives both $\operatorname{coker} \epsilon = 0$ and the short exact sequence

$$0 \longrightarrow \ker \epsilon' \longrightarrow \ker \epsilon \longrightarrow \ker \epsilon'' \longrightarrow 0.$$

The notes also check surjectivity directly for modules: given $a \in A$, choose x'' with $\epsilon''(x'') = \pi(a)$. Then $a - \beta(x'') \in \ker \pi = \operatorname{im} i$, and surjectivity of ϵ' gives x' with $a - \beta(x'') = i\epsilon'(x')$. Thus $a = \epsilon(x', x'')$. The snake-lemma argument is the categorical version and does not require elements of arbitrary abelian-category objects.

Repeat the same construction on the short exact sequence of kernels, using $P'_1 \twoheadrightarrow \ker \epsilon'$ and $P''_1 \twoheadrightarrow \ker \epsilon''$. The resulting epimorphism $P'_1 \oplus P''_1 \to \ker \epsilon$, followed by the kernel inclusion into P_0, defines d_1. Iteration constructs d_n and makes $\operatorname{im} d_{n+1} = \ker d_n$ at every degree. Finite direct sums of projectives are projective, completing the construction.

The off-diagonal differential. Relative to the chosen biproducts, the differential has the form

$$d_n^P = \begin{pmatrix} d'_n & \alpha_n \\ 0 & d''_n \end{pmatrix}, \qquad \alpha_n : P''_n \longrightarrow P'_{n-1}. \tag{6.2}$$

The equation $(d^P)^2 = 0$ says $d'_{n-1}\alpha_n + \alpha_{n-1}d''_n = 0$ for $n \geq 2$; in degree one the augmentation condition is $i\epsilon'\alpha_1 + \beta d''_1 = 0$. The α_n encode how the two resolutions are joined. Thus the horseshoe strengthens comparison: it produces compatible lifts with an exact, degreewise split sequence of resolutions, rather than merely independent comparison maps.

The handwritten integer example. Consider $0 \to \mathbb{Z}/2 \xrightarrow{i} \mathbb{Z}/4 \xrightarrow{\pi} \mathbb{Z}/2 \to 0$, where $i([a]_2) = [2a]_4$ and $\pi([b]_4) = [b]_2$. Resolve each end by $0 \to \mathbb{Z} \xrightarrow{2} \mathbb{Z} \to \mathbb{Z}/2 \to 0$. Taking $\beta(1) = [1]_4$ yields $\epsilon(a,b) = [2a+b]_4$. Its kernel is $K = \{(a,b) \in \mathbb{Z}^2 : 2a + b \equiv 0 \pmod 4\}$, and $0 \to \mathbb{Z} \to K \to \mathbb{Z} \to 0$ is exact, with first map $x \mapsto (2x, 0)$ and second map $(a,b) \mapsto b/2$. Equivalently, before identifying the right kernel $2\mathbb{Z}$ with $\mathbb{Z}$, the second map is $(a,b) \mapsto b$.

To lift the right differential 2, choose a vector $(c,2) \in K$. The condition $2c + 2 \equiv 0 \pmod 4$ permits $c = -1$. Hence the horseshoe resolution is

$$0 \longrightarrow \mathbb{Z}^2 \xrightarrow{D} \mathbb{Z}^2 \xrightarrow{\epsilon} \mathbb{Z}/4 \longrightarrow 0, \qquad D = \begin{pmatrix} 2 & -1 \\ 0 & 2 \end{pmatrix}.$$

Indeed $\epsilon D = 0$, D is injective, and every $(a,b) \in K$ has $b = 2y$ and $a + y = 2x$, so $(a,b) = D(x,y)$. The off-diagonal -1 records the nonsplit extension. For the split middle object $\mathbb{Z}/2 \oplus \mathbb{Z}/2$, the same choices give the diagonal matrix $\mathrm{diag}(2,2)$ and augmentation $(a,b) \mapsto ([a]_2, [b]_2)$, exactly as noted in blue in the manuscript.

7 Injectives, derived functors, and adjunctions

7.1 Injective objects and Baer's criterion

An object I of an abelian category is *injective* if, for every monomorphism $u : A \hookrightarrow B$ and every $a : A \to I$, there exists $b : B \to I$ with $bu = a$. Thus projectivity lifts a map through an epimorphism, whereas injectivity extends a map from a subobject. Figure 21 fixes the directions.

A product $\prod_\lambda I_\lambda$, whenever it exists, is injective if each I_λ is injective: extend every component separately and combine them using the product universal property. This includes infinite products, with the usual set-theoretic choice of extensions. The notes retain the historical observation that Baer's work on injective modules dates to 1940, before the systematic treatment of projectives by Cartan and Eilenberg in 1956; see the historical discussion in [Wei94, §2.3].

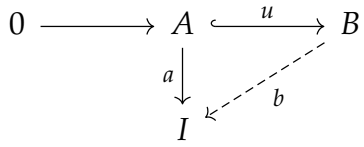


Figure 21. The extension property defining an injective object. The dashed extension need not be unique.

Equivalent criteria. The following conditions are equivalent: I is injective; the contravariant functor $\mathrm{Hom}_{\mathcal{A}}(-, I)$ is exact; every monomorphism $I \hookrightarrow Y$ splits. The first equivalence follows from left exactness of $\mathrm{Hom}(-, I)$ and the extension property, which provides the missing surjectivity. For the second, injectivity extends id_I to a retraction $r : Y \to I$, giving $Y \cong I \oplus \ker r$.

Conversely, take $u : A \hookrightarrow B$ and $f : A \to I$, and form the pushout Y of u along f. In a module category this is

$$Y = (I \oplus B)/\{(f(a), -u(a)) : a \in A\}.$$

The map $j : I \to Y$, $x \mapsto [(x, 0)]$, is monic: a relation $(x, 0) = (f(a), -u(a))$ forces $a = 0$. If every such monomorphism splits, choose $r : Y \to I$ with $rj = \mathrm{id}_I$. Then $\widetilde{f}(b) = r([(0, b)])$ extends f, since $[(0, u(a))] = [(f(a), 0)]$. In an arbitrary abelian category the same proof uses the cokernel defining the pushout and the fact that the pushout of a monomorphism is monic.

Baer's criterion. A right R-module E is injective if and only if every homomorphism $J \to E$ from a right ideal $J \subseteq R$ extends to $R \to E$. Necessity follows immediately from the definition. For sufficiency, given $A \subseteq B$ and $a : A \to E$, order its extensions $a' : A' \to E$, $A \subseteq A' \subseteq B$, by inclusion and agreement. The union of a chain gives a well-defined upper bound, so Zorn's lemma supplies a maximal extension.

If $A' \neq B$, choose $b \in B \setminus A'$ and put $J = \{r \in R : br \in A'\}$. This is a right ideal, and $r \mapsto a'(br)$ extends, by hypothesis, to an R-linear map $f : R \to E$. Define $a''(x + br) = a'(x) + f(r)$ on $A' + bR$. To check well-definedness, if $x + br = x' + br'$, then $b(r - r') = x' - x \in A'$, and consequently $f(r - r') = a'(x' - x)$. Hence both expressions for a'' agree. This contradicts maximality. Therefore $A' = B$, proving injectivity.

Divisibility and the examples in the notes. Over a principal ideal domain—a commutative integral domain whose ideals are each generated by one element—an R-module E is injective exactly when it is *divisible*: for each nonzero $r \in R$, multiplication by r is onto E. To see the link with Baer's criterion, a map $rR \to E$ is determined by the image x of r; its extension to R amounts to finding y with $yr = x$. The zero ideal poses no condition. For $R = \mathbb{Z}$, this says that each equation $ny = x$ with $n \geq 1$ is soluble.

Thus $\mathbb{Z}$ is not injective, but $\mathbb{Q}$ and the Prüfer group $\mathbb{Z}_{p^\infty} = \mathbb{Z}[1/p]/\mathbb{Z}$ are injective. The latter is the increasing union of $p^{-k}\mathbb{Z}/\mathbb{Z}$, whose elements are $0, 1/p^k, \dots, (p^k - 1)/p^k$ modulo $\mathbb{Z}$. Multiplication by p is onto by moving to the next stage, and multiplication by an integer coprime to p is an automorphism at every finite stage. This proves divisibility. *Editorial correction:* the endpoint of the finite list is $(p^k - 1)/p^k$, rather than the ambiguous handwritten exponent in that position.

Structure of divisible abelian groups. The manuscript records the structure theorem

$$D \cong \mathbb{Q}^{(\kappa_0)} \oplus \bigoplus_{p \text{ prime}} \mathbb{Z}_{p^\infty}^{(\kappa_p)}$$

for a divisible group D; parentheses denote direct sums indexed by cardinals. This is the standard theorem cited by [Wei94, Example 2.3.3], rather than a new conclusion of the preceding finite examples. The notes' phrase "split the torsion part" is a useful proof outline: its torsion subgroup is divisible, hence injective, and splits off; the torsion-free quotient is a $\mathbb{Q}$-vector space. The torsion summand decomposes into its p-primary parts, and the structure theorem for divisible p-primary groups identifies each with a sum of Prüfer groups. No classification proof is supplied in the handwriting. In the special case needed below, every rational class has finite order, and the decomposition of its order into prime powers gives

$$\mathbb{Q}/\mathbb{Z} \cong \bigoplus_{p \text{ prime}} \mathbb{Z}_{p^\infty}.$$

This particular isomorphism is the primary decomposition furnished by the Chinese remainder theorem; it does not by itself prove the full classification theorem.

Enough injectives. For an abelian group A, let $S = \mathrm{Hom}_{\mathbf{Ab}}(A, \mathbb{Q}/\mathbb{Z})$ and $I(A) = \prod_{f \in S} \mathbb{Q}/\mathbb{Z}$. This product is injective, and the evaluation map $e_A : A \to I(A)$ sends a to $(f(a))_{f \in S}$. To prove it injective, take $a \neq 0$. If a has finite order n, define a map on $\langle a \rangle$ by $ka \mapsto k/n + \mathbb{Z}$. If a has infinite order, use $ka \mapsto k/2 + \mathbb{Z}$. In either case the image of a is nonzero. Injectivity of $\mathbb{Q}/\mathbb{Z}$ extends this map to all of A, giving a nonzero coordinate of $e_A(a)$. Thus every abelian group embeds in an injective, and $A = 0$ if and only if $\mathrm{Hom}_{\mathbf{Ab}}(A, \mathbb{Q}/\mathbb{Z}) = 0$.

7.2 Coinduction and adjoint functors

For an abelian group V, give $\mathrm{Hom}_{\mathbf{Ab}}(R, V)$ the right R-action $(\phi \cdot k)(r) = \phi(kr)$. The order kr is essential: it makes $((\phi \cdot k) \cdot \ell)(r) = \phi(k\ell r)$. For any right R-module M, there is a natural isomorphism

$$\tau : \mathrm{Hom}_{\mathbf{Ab}}(M, V) \longrightarrow \mathrm{Hom}_R(M, \mathrm{Hom}_{\mathbf{Ab}}(R, V)), \qquad (\tau f)(m)(r) = f(mr).$$

Its inverse is evaluation at 1: $\mu(g)(m) = g(m)(1)$. Indeed R-linearity gives $g(mr)(1) = (g(m)\cdot r)(1) = g(m)(r)$, so $\tau\mu(g) = g$; and $\mu\tau(f)(m) = f(m1) = f(m)$. This provides a concrete model for adjunction before introducing its abstract definition.

The natural Hom correspondence. Functors $L : \mathcal{A} \to \mathcal{B}$ and $R : \mathcal{B} \to \mathcal{A}$ are *adjoint*, written $L \dashv R$, if there are bijections $\tau_{A,B} : \mathrm{Hom}_{\mathcal{B}}(LA, B) \cong \mathrm{Hom}_{\mathcal{A}}(A, RB)$ natural in both variables. The first variable is contravariant and the second covariant: for $u : A \to A'$ and $v : B \to B'$, naturality means

$$\tau_{A,B}(h \circ Lu) = \tau_{A',B}(h) \circ u, \qquad \tau_{A,B'}(v \circ k) = Rv \circ \tau_{A,B}(k).$$

These identities are drawn in Figure 22. The preceding construction says that the forgetful functor $U : \mathsf{Mod}\text{-}R \to \mathbf{Ab}$ has right adjoint $\mathrm{Hom}_{\mathbf{Ab}}(R, -)$, often called coinduction.

$$\begin{array}{ccccc}
\mathrm{Hom}_{\mathcal{B}}(LA', B) & \xrightarrow{(Lu)^*} & \mathrm{Hom}_{\mathcal{B}}(LA, B) & \xrightarrow{v_*} & \mathrm{Hom}_{\mathcal{B}}(LA, B') \\
\downarrow{\scriptstyle \tau} & & \downarrow{\scriptstyle \tau} & & \downarrow{\scriptstyle \tau} \\
\mathrm{Hom}_{\mathcal{A}}(A', RB) & \xrightarrow[u^*]{} & \mathrm{Hom}_{\mathcal{A}}(A, RB) & \xrightarrow[(Rv)_*]{} & \mathrm{Hom}_{\mathcal{A}}(A, RB')
\end{array}$$

Figure 22. Naturality of the adjunction correspondence. A superscript star denotes precomposition, and a subscript star denotes postcomposition.

Preserving injectives and projectives. Suppose $L \dashv R$ between abelian categories and L preserves monomorphisms, for example if L is left exact. Then R sends injectives to injectives. Given $u : A \hookrightarrow A'$ and I injective in $\mathcal{B}$, the restriction map $\mathrm{Hom}_{\mathcal{B}}(LA', I) \to \mathrm{Hom}_{\mathcal{B}}(LA, I)$ is onto because Lu is monic. Under the natural adjunction bijections it becomes $\mathrm{Hom}_{\mathcal{A}}(A', RI) \to \mathrm{Hom}_{\mathcal{A}}(A, RI)$, proving the extension property for RI. Dually, if R preserves epimorphisms, then L preserves projectives. The exactness hypotheses stated in [Wei94, Proposition 2.3.10] imply these sufficient conditions; the manuscript records the weaker one-sided conditions used by the proof.

In particular, since the forgetful functor U is exact, $\mathrm{Hom}_{\mathbf{Ab}}(R, I)$ is an injective right R-module when I is an injective abelian group. *Editorial completion of the construction:* choose a monomorphism $UM \hookrightarrow I$ in $\mathbf{Ab}$. Its adjoint $M \to \mathrm{Hom}_{\mathbf{Ab}}(R, I)$ is monic because evaluation at 1 recovers the chosen monomorphism. Hence module categories have enough injectives too. Iterating embeddings of successive cokernels yields an injective resolution $0 \to M \to I^0 \to I^1 \to \cdots$; this is the dual construction to projective resolutions.

7.3 Left derived functors

Let $F : \mathcal{A} \to \mathcal{B}$ be an additive right exact functor between abelian categories, and suppose $\mathcal{A}$ has enough projectives. Choose a projective resolution $P_\bullet \to A$ and define

$$L_nF(A) = H_n(F(P_\bullet)), \qquad n \geq 0. \tag{7.1}$$

Applying F term by term preserves the relation $d^2 = 0$, but may destroy exactness. The resulting homology therefore measures the failure of F to preserve the exact resolution. Right exactness gives $F(P_1) \to F(P_0) \to F(A) \to 0$ exact, so $L_0F(A) \cong F(A)$.

If $Q_\bullet \to A$ is another projective resolution, comparison gives a homotopy equivalence $P_\bullet \simeq Q_\bullet$. An additive functor preserves the homotopy equation, so it induces a canonical isomorphism $H_n(F(P)) \cong H_n(F(Q))$ independent of the comparison lift. For a morphism $f : A' \to A$, choose a comparison lift $\widetilde{f} : P'_\bullet \to P_\bullet$ and define $L_nF(f) = H_n(F(\widetilde{f}))$. Homotopy uniqueness makes this well-defined; lifts of identities, compositions, and sums prove functoriality and additivity. Different simultaneous choices of resolutions give naturally isomorphic functors, not literally identical chosen complexes.

If A is projective, use the resolution with $P_0 = A$, augmentation id_A, and all positive terms zero; then $L_nF(A) = 0$ for $n > 0$. An object Q is called *F-acyclic* if $L_nF(Q) = 0$ for every $n > 0$. The manuscript previews the acyclic-resolution theorem: if an exact nonnegative resolution $Q_\bullet \to A$ has F-acyclic terms, then $H_n(F(Q_\bullet)) \cong L_nF(A)$. This is retained as a stated result, with its proof deferred to [Wei94, F-Acyclic Objects 2.4.3 and Exercise 2.4.3]. It does not follow from the definition of acyclicity alone; the exercise proves it by successive short exact sequences of kernels and their connecting maps. The constructions below use projective resolutions and do not assume this deferred proof. The projective case is the basic example, not the definition of acyclicity.

Exact postcomposition: the handwritten solution of Exercise 2.4.2. If $U : \mathcal{B} \to \mathcal{C}$ is additive and exact, then $U(L_nF) \cong L_n(UF)$. For a complex C, write $Z_n = \ker d_n$ and $B_n = \operatorname{im} d_{n+1}$. Apply U to the short exact sequences $0 \to Z_n \to C_n \to B_{n-1} \to 0$ and $0 \to B_n \to Z_n \to H_n(C) \to 0$. Exactness identifies cycles and boundaries of $U(C)$ with $U(Z_n)$ and $U(B_n)$ and therefore gives $H_n(U(C)) \cong U(H_n(C))$. Taking $C = F(P_\bullet)$ proves the formula. The notes point out that exact forgetful functors often simplify a calculation by discarding extra structure without changing the underlying homology.

Constructing the connecting morphisms. For a short exact sequence $0 \to A' \to A \to A'' \to 0$, choose a horseshoe $0 \to P'_\bullet \to P_\bullet \to P''_\bullet \to 0$. Since each degree is split, applying the additive functor F leaves a short exact sequence of complexes, even though F need not be left exact. Its homology connecting maps define $\delta_n : L_nF(A'') \to L_{n-1}F(A')$ and the long exact sequence. The construction must be independent of the

chosen horseshoe and natural in short exact sequences. The manuscript supplies the following detailed lifting argument rather than leaving this compatibility implicit.

Naturality: lifting a morphism of horseshoes. Consider a commutative diagram of short exact sequences with vertical maps $f' : A' \to B'$, $f : A \to B$, and $f'' : A'' \to B''$. Choose horseshoes P and Q, and comparison lifts $u' : P' \to Q'$ and $u'' : P'' \to Q''$. We seek a compatible chain map $u : P \to Q$ of the form

$$u_n = \begin{pmatrix} u'_n & \gamma_n \\ 0 & u''_n \end{pmatrix}, \qquad \gamma_n : P''_n \longrightarrow Q'_n.$$

Editorial correction: the source of γ_n must be the double-primed summand; this follows from its position in the matrix and repairs the single-prime slip in the handwriting. Write $\lambda_P : P''_0 \to A$ and $\lambda_Q : Q''_0 \to B$ for the chosen augmentation lifts. Degree-zero compatibility requires

$$i_B \eta' \gamma_0 = f \lambda_P - \lambda_Q u''_0.$$

Applying π_B to the right side gives $f'' \epsilon'' - \eta'' u''_0 = 0$. It therefore factors through $i_B : B' \hookrightarrow B$ as a map $b : P''_0 \to B'$. Projectivity lifts b through $\eta' : Q'_0 \twoheadrightarrow B'$, giving γ_0.

For $n \geq 1$, write the off-diagonal differentials as α^P_n and α^Q_n, as in (6.2). Having defined $\gamma_0, \dots, \gamma_{n-1}$, the chain-map equation reduces to

$$d^{Q'}_n \gamma_n = g_n, \qquad g_n = \gamma_{n-1} d^{P''}_n - \alpha^Q_n u''_n + u'_{n-1} \alpha^P_n. \tag{7.2}$$

The source stresses that one must show g_n lands in the cycles *before* constructing γ_n; using the equation to be solved would be circular. For $n \geq 2$, expansion using the already established equation for γ_{n-1} gives

$$d^{Q'}_{n-1} g_n = u'_{n-2}(\alpha^P_{n-1} d^{P''}_n + d^{P'}_{n-1} \alpha^P_n) - (\alpha^Q_{n-1} d^{Q''}_n + d^{Q'}_{n-1} \alpha^Q_n) u''_n = 0.$$

Both parentheses vanish by the horseshoe differential identity. For $n = 1$, use η' instead of $d^{Q'}_0$: applying i_B to $\eta' g_1$ gives

$$i_B \eta' g_1 = (f \lambda_P - \lambda_Q u''_0) d^{P''}_1 + \lambda_Q d^{Q''}_1 u''_1 + f i_A \epsilon' \alpha^P_1.$$

The two terms containing λ_Q cancel because u'' is a chain map, leaving $f(\lambda_P d^{P''}_1 + i_A \epsilon' \alpha^P_1) = 0$. Since i_B is monic, $\eta' g_1 = 0$. Thus in every degree g_n factors through the appropriate kernel, and projectivity of P''_n lifts it to γ_n.

The resulting map of short exact sequences of complexes, shown in Figure 23, makes the homology connecting maps commute. Taking identity maps on the original short exact sequence also proves independence from the chosen resolutions and horseshoes. Therefore

$$L_{n-1}F(f')\, \delta^A_n = \delta^B_n\, L_n F(f'').$$

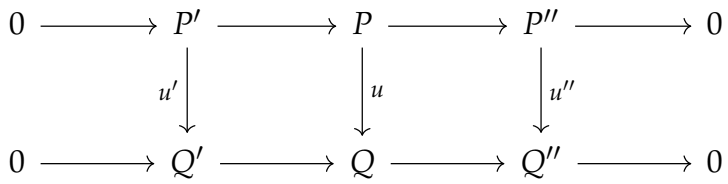


Figure 23. Compatible comparison maps between horseshoes. All arrows are chain maps and both rows are degreewise split exact.

7.4 Universality

The convention in the manuscript agrees with [Wei94, Definition 2.1.4]: a homological δ-functor S_* is universal when every natural transformation $T_0 \to S_0$ from another homological δ-functor extends uniquely to a morphism $T_* \to S_*$. Thus universality of left derived functors means that $\varphi_0 : T_0 \to F$ extends uniquely to $\varphi_n : T_n \to L_nF$. The direction matters; the dual cohomological convention reverses it.

The proof in the notes proceeds in three stages: construct each component using a projective presentation, prove naturality and independence of that presentation, and then compare an arbitrary short exact sequence with such a presentation. Suppose φ_i has been constructed for $i < n$. Choose $0 \to K \xrightarrow{j} P \to A \to 0$ with P projective. Since $L_nF(P) = 0$, the lower row of Figure 24 identifies δ^L_n with a kernel. Naturality of φ_{n-1} gives

$$L_{n-1}F(j)\varphi^K_{n-1}\delta^T_n = \varphi^P_{n-1}T_{n-1}(j)\delta^T_n = 0.$$

The kernel universal property therefore gives a unique φ^A_n satisfying $\delta^L_n\varphi^A_n = \varphi^K_{n-1}\delta^T_n$.

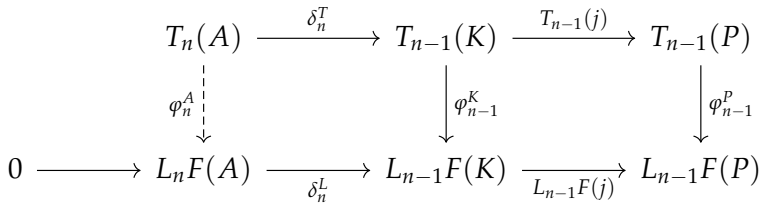


Figure 24. The inductive construction of the universal transformation. The displayed portions of both rows are exact, and the lower connecting map is monic.

Given $a : A' \to A$ and another presentation $0 \to K' \to P' \to A' \to 0$, lift a to $b : P' \to P$ and let $c : K' \to K$ be its restriction. The kernel property defines c uniquely. Naturality of the two connecting maps and the induction hypothesis then give

$$\delta^L_n L_nF(a)\varphi^{A'}_n = L_{n-1}F(c)\varphi^{K'}_{n-1}\delta^T_n = \varphi^K_{n-1}T_{n-1}(c)\delta^T_n = \delta^L_n\varphi^A_n T_n(a).$$

Cancel the monomorphism δ^L_n to obtain naturality. Taking $a = \mathrm{id}_A$ and different projective presentations proves independence of the choice. Figure 25 redraws the small cube in the handwritten page as the comparison of two composites; its top connecting maps point from T_n to T_{n-1}, while the lower connecting maps point from L_nF to $L_{n-1}F$.

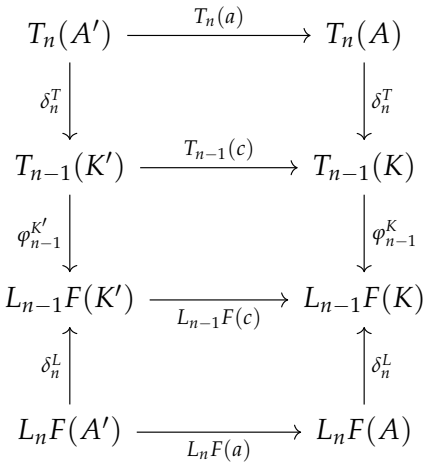


Figure 25. The commutative faces used in the universality proof. By construction, the composite from each top object to the third row also equals $\delta_n^L \varphi_n$. Canceling the lower-right monomorphism proves the missing outer naturality square.

Finally, for $0 \to A' \to A \to A'' \to 0$, choose $0 \to K'' \to P'' \to A'' \to 0$ with P'' projective. Lift $P'' \to A''$ to $P'' \to A$, inducing $g : K'' \to A'$. Naturality of connecting maps identifies the connecting map of the original sequence with the composite through K''. The diagram in Figure 26 commutes by the construction of φ_n and the established naturality of φ_{n-1}, so φ respects the connecting map of every short exact sequence. Uniqueness holds inductively because the kernel factorization in Figure 24 is unique.

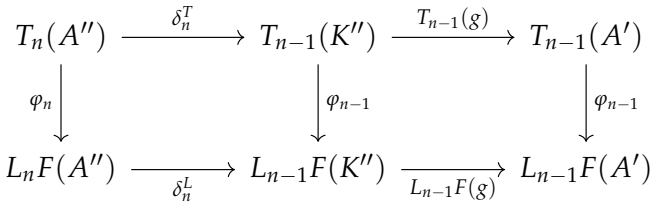


Figure 26. Compatibility with the connecting maps of an arbitrary short exact sequence. Here each horizontal composite, rather than its first arrow alone, is the connecting map for that original sequence.

The retained effaceability exercise. The manuscript includes [Wei94, Exercise 2.4.5]. An additive functor G is *effaceable* if every object A admits a monomorphism $u : A \to I$ with $G(u) = 0$, and *coeffaceable* if every A admits an epimorphism $u : P \to A$ with $G(u) = 0$. The exercise asks one to prove that a homological δ-functor T_* is universal when each T_n, $n > 0$, is coeffaceable, and dually that a cohomological δ-functor is universal when each positive-degree functor is effaceable.

The notes explicitly leave this proof unfinished and record the correct replacement for projective lifting: use a pullback. More precisely, for $u : P \to Z$ and $p : Y \to Z$,

a lift $\ell : P \to Y$ with $p\ell = u$ is equivalent to a section of $P \times_Z Y \to P$. The pullback produces an epimorphism to P when p is epic; it does not automatically produce a section. For coeffaceability, a common pullback refines two choices of epimorphisms without assuming either source projective. Figure 27 reconstructs the tiny pasted diagram and this logical distinction. The full exercise is retained as an exercise, rather than attributed to the handwritten proof.

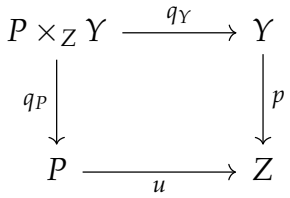


Figure 27. The pullback relation $pq_Y = uq_P$. A section s of q_P yields a lift $q_Y s$; a lift conversely determines such a section by the pullback universal property.

7.5 Right derived functors, Ext, and Tor

Let $F : \mathcal{A} \to \mathcal{B}$ be additive and left exact, and assume $\mathcal{A}$ has enough injectives. Choose $0 \to A \to I^0 \to I^1 \to \cdots$ and define $R^n F(A) = H^n(F(I^\bullet))$. Left exactness identifies $R^0 F(A)$ with $F(A)$. Passing to opposite categories turns F into a right exact functor $F^{\mathrm{op}} : \mathcal{A}^{\mathrm{op}} \to \mathcal{B}^{\mathrm{op}}$, so the preceding results dualize: choices yield canonical natural isomorphisms, injectives have vanishing positive derived functors, and R^*F is a universal cohomological δ-functor. Under this convention a transformation $F \to T^0$ extends uniquely to $R^*F \to T^*$. A right F-acyclic object Q satisfies $R^n F(Q) = 0$ for all $n > 0$; an exact sequence $0 \to A \to Q^0 \to Q^1 \to \cdots$ with such F-acyclic terms computes R^*F by applying F and taking cohomology. This last acyclic-resolution assertion is the dual of the stated, proof-deferred result above, as recorded on handwritten page 32 and in [Wei94, Section 2.5]; it is not a separate derivation here.

Contravariant functors. A contravariant left exact functor $F : \mathcal{A} \to \mathcal{B}$ is an ordinary left exact functor $\widetilde{F} : \mathcal{A}^{\mathrm{op}} \to \mathcal{B}$. It takes $0 \to A' \to A \to A'' \to 0$ to $0 \to F(A'') \to F(A) \to F(A')$. If $\mathcal{A}$ has enough projectives, these are injectives in $\mathcal{A}^{\mathrm{op}}$, so a projective resolution $P_\bullet \to A$ computes

$$R^n F(A) = H^n(F(P_\bullet)).$$

Here $F(P_\bullet)$ is a cochain complex: $F(d_{n+1}) : F(P_n) \to F(P_{n+1})$. *Editorial correction:* the handwritten formula $H^n(P_\bullet)$ omits the application of F; its intended, type-correct form is the displayed expression.

The two basic bifunctors and the retained exercises. For right R-modules A, B, define $\operatorname{Ext}^n_R(A,B) = R^n \operatorname{Hom}_R(A,-)(B)$. In other words, resolve the second argument injectively and take cohomology of $\operatorname{Hom}_R(A, I^\bullet)$. The notes then retain [Wei94, Exercises 2.5.1–2.5.2]: B is injective if and only if $\operatorname{Hom}_R(-,B)$ is exact, if and only if $\operatorname{Ext}^n_R(A,B) = 0$ for all A and $n > 0$, if and only if $\operatorname{Ext}^1_R(A,B) = 0$ for all A. Dually, A is projective if and only if $\operatorname{Hom}_R(A,-)$ is exact, if and only if $\operatorname{Ext}^n_R(A,B) = 0$ for all B and $n > 0$, if and only if $\operatorname{Ext}^1_R(A,B) = 0$ for all B. These are exercise statements in the source; their solutions are not part of its written derivation. The quantifiers "for all A" and "for all B" are essential.

For a right R-module A and a left R-module B, the right exact functor $- \otimes_R B$ defines $\operatorname{Tor}^R_n(A,B) = L_n(- \otimes_R B)(A)$. Thus Tor is computed by applying $- \otimes_R B$ to a projective resolution of A and taking homology. Without additional bimodule structure its values are abelian groups. This explicitly fixes the sidedness suppressed in the handwritten formula.

7.6 Adjunction and one-sided exactness

The notes return to adjunctions to explain a source of exactness. For additive adjoint functors $L : \mathcal{A} \rightleftarrows \mathcal{B} : R$ between abelian categories, R is left exact and L is right exact. Given $0 \to B' \to B \to B'' \to 0$, apply $\operatorname{Hom}_{\mathcal{B}}(LA, -)$ and transport the resulting exact sequence across the natural adjunction isomorphisms. For every A this gives

$$0 \longrightarrow \operatorname{Hom}_{\mathcal{A}}(A, RB') \longrightarrow \operatorname{Hom}_{\mathcal{A}}(A, RB) \longrightarrow \operatorname{Hom}_{\mathcal{A}}(A, RB'').$$

The naturality squares are those of Figure 28. *Editorial reconstruction of the Yoneda step:* the first map is injective for every A, so $RB' \to RB$ is monic. Every $a : A \to RB$ whose composite into RB'' vanishes has a unique factorization through RB'. That is precisely the universal property of the kernel. Hence $0 \to RB' \to RB \to RB''$ is exact. The opposite-category argument proves right exactness of L.

$$\begin{array}{ccccccc}
0 & \longrightarrow & \operatorname{Hom}_{\mathcal{B}}(LA, B') & \longrightarrow & \operatorname{Hom}_{\mathcal{B}}(LA, B) & \longrightarrow & \operatorname{Hom}_{\mathcal{B}}(LA, B'') \\
 & & \downarrow \cong & & \downarrow \cong & & \downarrow \cong \\
0 & \longrightarrow & \operatorname{Hom}_{\mathcal{A}}(A, RB') & \longrightarrow & \operatorname{Hom}_{\mathcal{A}}(A, RB) & \longrightarrow & \operatorname{Hom}_{\mathcal{A}}(A, RB'')
\end{array}$$

Figure 28. Transporting left exactness through an adjunction. Each square commutes by naturality in the second variable; neither row claims surjectivity at its last term.

The unit as a universal factorization. Define $\eta_A = \tau_{A,LA}(\mathrm{id}_{LA}) : A \to RLA$, called the *unit* of the adjunction. Every $\alpha : A \to RB$ has a unique correspondent $g : LA \to B$.

Naturality in B gives $\alpha = \tau_{A,B}(g \circ \mathrm{id}_{LA}) = R(g)\eta_A$. Figure 29 displays this factorization. Uniqueness is uniqueness of g in $\mathcal{B}$; the diagram's label $R(g)$ does not claim that R is faithful on arbitrary morphisms.

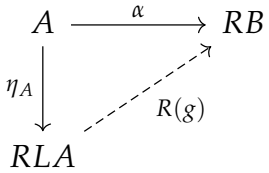


Figure 29. The universal arrow from A to the right adjoint R. There is a unique $g : LA \to B$ inducing the indicated factorization.

Constructing the left adjoint from universal arrows. Fix $R : \mathcal{B} \to \mathcal{A}$. A left adjoint exists if and only if, for every A, one can choose B_A and $\eta_A : A \to R(B_A)$ such that each $\alpha : A \to R(B)$ factors uniquely as $R(h)\eta_A$ for a morphism $h : B_A \to B$. Necessity was just proved. Conversely, set $L(A) = B_A$. For $u : A \to A'$, apply the universal property to $\eta_{A'}u$ to obtain the unique $L(u) : B_A \to B_{A'}$ with $R(L(u))\eta_A = \eta_{A'}u$. Uniqueness then gives $L(\mathrm{id}_A) = \mathrm{id}_{B_A}$ and $L(vu) = L(v)L(u)$, so this construction really is a functor.

Finally define $\tau_{A,B}(h) = R(h)\eta_A$. Existence of factorization makes this map surjective, and uniqueness makes it injective. Naturality in B follows from functoriality of R, and naturality in A follows from the defining equation for $L(u)$. This finishes the converse, including the functoriality and naturality checks abbreviated in the source.

The final application: tensor–Hom adjunction. The final handwritten page applies this mechanism to a left R-module B: the functor $- \otimes_R B : \mathsf{Mod}\text{-}R \to \mathbf{Ab}$ is left adjoint to $\mathrm{Hom}_{\mathbf{Ab}}(B, -)$. More generally, let ${}_R B_S$ be an R–S bimodule: B has a left R-action and a right S-action satisfying $(rb)s = r(bs)$. For a right R-module M and right S-module N, there is a natural isomorphism

$$\mathrm{Hom}_S(M \otimes_R B, N) \cong \mathrm{Hom}_R(M, \mathrm{Hom}_S(B, N)),$$

where the right R-action on $\mathrm{Hom}_S(B, N)$ is $(f \cdot r)(b) = f(rb)$. *Editorial expansion of the stated application:* a map h on the tensor product corresponds to $m \mapsto (b \mapsto h(m \otimes b))$. The inverse sends g to $m \otimes b \mapsto g(m)(b)$; the equation $g(mr)(b) = g(m)(rb)$ is exactly what makes this inverse balanced over R. These formulas prove the adjunction and explain, through the preceding theorem, the right exactness of tensor product and left exactness of the corresponding Hom functor.

A Source correspondence and editorial decisions

The manuscript's progression is preserved. Pages 1–4 develop additive and abelian categories; pages 5–9 treat the snake lemma and long exact sequences; pages 10–14

discuss chain homotopies, cones and cylinders; pages 15–18 introduce functor categories, Yoneda and an unfinished embedding argument. Pages 19–20 introduce δ-functors, pages 20–25 develop projectives and resolutions, pages 26–28 develop injectives, and pages 28–34 construct derived functors and discuss adjunctions. A proof crossing a page boundary is presented continuously. The machine-readable coverage ledger gives finer, labelled correspondences for individual source units.

Preserving the author's choices. The signs of the mapping cone and its quotient map are retained together. In particular, the manuscript uses $d(b,c) = (-d_B b, d_C c - fb)$ and the quotient $(b,c) \mapsto -b$. Changing only one of these signs would change the connecting map in the displayed long exact sequence. The redrawn diagrams therefore follow the actual handwritten convention, with their commutative relations stated in the surrounding prose. Repeated diagrams expressing the same relation may be combined, but their mathematically distinct maps and claims are retained.

Corrections that affect mathematical meaning. On manuscript page 9, the sequence built from the cycle and boundary complexes has connecting map $B_{n-1}(C) \hookrightarrow Z_{n-1}(C)$, not the zero map. The proposed short exact sequence involving $Z_n(C)$, $H_n(C)$ and $B_{n-1}(C)$ is therefore replaced by the correct calculation and an explicit diagnostic example. On page 11, the statement about the homotopy category failing to be abelian is interpreted as a statement about the general situation, rather than every possible coefficient category.

The Freyd–Mitchell material on pages 17–18 is retained as an incomplete outline, as indicated by the author. Its established Yoneda and weak-effacement arguments are separated from the larger embedding theorem. The published correction to Weibel's page 29 concerns an embedding of a small full abelian subcategory into modules; the overstrong assertion that the entire auxiliary category is equivalent to a module category is not adopted [Wei]. The unfinished construction of a suitable projective object is not silently supplied or claimed to be proved here.

In the horseshoe naturality argument on manuscript page 29, the off-diagonal correction has the domain dictated by the second summand of the source resolution. On page 32, the inverse-variance derived-functor formula must apply F to the projective resolution before taking cohomology. These repairs are explained at the corresponding passages. Ordinary OCR mistakes, such as confusing a category symbol with the empty set, duplicating arrows, or losing primes and subscripts, have been normalized from the handwritten page rather than preserved as mathematical claims.

Supplements and verification boundary. Editorial additions supply the local definitions, missing factorization steps, sign checks and examples needed to make the original route readable. They do not replace the manuscript with a general textbook

survey. Source exercises remain identifiable even when a solution or a short verification is added. The unfinished Freyd–Mitchell outline remains the principal explicitly retained limitation. The mathematics is checked by analytic reasoning and direct comparison of sources and diagrams; no computer-algebra verification is claimed.

References

[Wei] Charles A. Weibel. Corrections to An Introduction to Homological Algebra. Author's errata for the 1995 paperback edition, https://sites.math.rutgers.edu/~weibel/Hbook-corrections.html. Consulted 7 November 2026; p. 29 correction.

[Wei94] Charles A. Weibel. *An Introduction to Homological Algebra*, volume 38 of *Cambridge Studies in Advanced Mathematics*. Cambridge University Press, 1994.